

Appendix

For Online Publication

Online Appendix Section [A](#) contains additional tables and figures referenced in the main text.

Section [A.1](#) contains additional result tables. Table [A1](#) shows that white-collar vacancies are much more likely to be posted formally. Table [A2](#) shows correlates of formal and informal job search based on labor force data. Table [A3](#) reports the accounting cost of the vacancy-posting subsidy. Table [A4](#) tests for balance between the treatment and control groups. Table [A5](#) tests for differential attrition by treatment groups. Table [A6](#) shows that firms who make use of the vacancy posting subsidy work in different sectors and differ on further characteristics. Table [A7](#) shows that vacancy posting subsidies led to a substitution away from other search channels. Table [A8](#) shows that the intervention had no measurable effect on the characteristics of new hires. Table [A9](#) reports mean search inputs for baseline vacancies, split by posting channel.

Table [A10](#) displays treatment effects on managers' beliefs. Table [A11](#) reports treatment effects on the firm-level number of applicants per vacancy. Table [A12](#) shows wage expectations and realized wages for different vacancy types. Table [A13](#) reports treatment heterogeneity by the pre-treatment hiring professionalism index. Table [A14](#) displays heterogeneous treatment effects by a wide range of observable characteristics. Table [A15](#) shows that effects on beliefs about applicant quality are driven by firms without experience of using formal search channels. Table [A16](#) shows that firms with experience in using formal hiring channels are larger and pay higher wages. Tables [A17](#) to [A19](#) show the additional impact of the screening intervention on vacancy posting and creation, and hiring outcomes.

Table [A20](#) compares three groups of job adverts: treated non-template postings, treated template postings, and formal postings observed among control-group firms. Table [A21](#) shows that the increase in white-collar va-

cancies is driven by vacancies to replace workers who left. Table A22 shows the impact on search inputs. Table A23 shows the impact on average salaries at endline. Table A24 displays treatment effects on turnover of existing employees. Table A25 shows treatment effects on firms' endline employee numbers and collar shares. Table A26 shows heterogeneous treatment effects on post-treatment behavior by baseline usage of formal search channels. Tables A27 and A28 show how take-up and treatment effects vary during the treatment period. Table A29 presents effects on firms' willingness to pay for the services offered as part of the treatments. Table A30 shows the impact on downstream business outcomes.

Section A.2 contains the main treatment effects estimates with control variables (Tables A31 to A39).

Section A.3 contains additional figures. Figure A1 shows the geographical distribution of firms in our sample. Figure A2 shows that managers perceive formal search as expensive and most suitable for white-collar workers. Figure A3 compares the cost of a formal vacancy posting to firms' baseline hiring costs and average salary levels. Figure A4 displays the timeline of the experiment. Figure A5 shows control group firm expectations about the relative quality of formal applicants compared to candidates obtained through social networks. Figure A6 shows control group firm expectations about the number of applicants that can be obtained through different formal posting channels. Figure A7 shows the difference between the actual and expected number of applicants by collar type for control group firms. Figure A8 displays the first page of a sample candidate skill report provided to firms in the subsidy plus screening group. Figure A9 compares formal and informal search by firms and job-seekers in Ghana and Vietnam. Figure A10 compares formal and informal search by firms and job-seekers in Malaysia.

Section B compares our sample to more representative data sources using Table B1.

Section C shows differences between white-collar and non-white-collar vacancies based on publicly posted vacancy data using Table C1.

In Section D we use Table D1 to explore how the differences between

wage expectations and realized earnings differ between job-seekers who search formally and those who do not.

Section [E](#) describes the details of our theoretical framework, which is introduced in Section [4](#).

Lastly, Section [F](#) lists the deviations from the pre-analysis plan.

A Additional exhibits

A.1 Additional tables

Table A1: Descriptive evidence on vacancy posting and occupational type

	Formal advertising	Informal advertising	# of vacancies (column %)
<i>Control group (row percentages):</i>			
Pooled	12%	88%	249 (100%)
White-collar	42%	58%	31 (12%)
Non-white-collar	8%	92%	218 (88%)

Notes: **Table A1 shows that white-collar vacancies are much more likely to be posted formally.** It displays the fraction of vacancies posted formally or informally by vacancy type (white-collar / non-white-collar) by firms in the control group of the experiment ($N = 216$). White-collar vacancies are defined as office and knowledge workers (e.g., managers, lawyers, administrators, accountants, and clerks).

Table A2: Correlates of formal and informal job search

	Formal search				Informal (network) search			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Years of education	0.01*** (0.00)			0.01*** (0.00)	0.00 (0.00)			0.00 (0.00)
First-time job-seekers		0.13*** (0.04)		0.08* (0.04)		-0.09* (0.05)		-0.06 (0.06)
Years of unemployment			0.02*** (0.01)	0.02*** (0.01)			-0.02* (0.01)	-0.02* (0.01)
Mean dep. var.	0.84	0.83	0.84	0.85	0.38	0.38	0.37	0.37
Observations	329	355	331	306	329	355	331	306

Notes: **Table A2 shows that education level, experience, and unemployment duration are correlated with formal search among job-seekers in Addis Ababa.** It presents correlates of formal and informal (network-based) job search. In columns 1-4, an indicator for whether a job-seeker searches in formal channels is regressed on her years of education, whether she is a first-time job-seeker, or both. Columns 5-8 repeat the same for informal job search as dependent variable. Based on data from the Ethiopian National Labor Force Survey 2021. Formal job search methods include online and offline job boards and newspapers. Informal job search methods are through social contacts/networks. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Cost of the vacancy-posting subsidy

	Firms	Vacancies	Mean	Median	p75	p90	Total
Posting cost per vacancy			5778.8	5778.8	5778.8	5778.8	
Cost per take-up firm	68	158	13427.1	5778.8	17336.2	28893.8	913042.5
Cost per assigned treatment firm	409	158	2232.4				913042.5
Position-weighted alternative	92	239	5778.8	5778.8	5778.8	5778.8	1381121.2

Notes: **Table A3 reports the accounting cost of the vacancy-posting subsidy.** Costs are reported in Ethiopian Birr (ETB); 100 ETB were worth around 5.21 USD PPP in 2019. The per-vacancy posting cost is the sum of newspaper, physical board, and online board prices used in the vacancy-cost figure. The take-up row reports total posting expenditures for firms that posted at least one subsidized vacancy. The assigned-firm row spreads the same total expenditure across all 409 firms assigned to treatment. The position-weighted alternative treats each requested worker slot as a separate posting and should therefore be interpreted as an upper-bound accounting exercise rather than the main subsidy measure.

Table A4: Treatment balance

	Control	Treatment	Δ	p(Control=Treatment)
Firm characteristics				
Age of firm (in years)	7.45	7.05	-0.404	0.548
# of employees	15.12	14.16	-0.952	0.352
Share of white-collar employees	0.13	0.15	0.014	0.271
Manufacturing sector	0.52	0.50	-0.024	0.563
Service sector (retail, hospitality)	0.27	0.28	0.008	0.836
Health Sector	0.07	0.13	0.060	0.013
Hiring practices				
Uses formal hiring channels	0.10	0.08	-0.021	0.391
Uses network hiring channels	0.81	0.79	-0.018	0.588
Uses other hiring channels	0.69	0.66	-0.027	0.488
Manager expectations				
Expected number of hires in next 3 months	3.06	3.67	0.618	0.159
Positive bus. outlook next 3 months	0.62	0.61	-0.008	0.840
Positive bus. outlook next 12 months	0.79	0.76	-0.028	0.441
Manager characteristics				
Female	0.30	0.23	-0.069	0.068
Manager age	34.98	35.50	0.519	0.565
Manager has univ. degree	0.42	0.47	0.051	0.226
Raven's Matrix score (out of 10)	5.01	4.83	-0.180	0.384

Notes: Table A4 shows that treatment and control groups have similar observable characteristics. It presents tests for equality of means across the treatment and control groups.

Table A5: Attrition analysis

	During treatment period				Post treatment period	
	(1) Any highfreq survey	(2) # highfreq surveys	(3) Has endline survey	(4) Has highfreq or endline survey	(5) Any post treatment survey	(6) # post treatment surveys
Treatment	-0.005 (0.017)	0.171 (0.193)	0.003 (0.015)	-0.010** (0.005)	0.010 (0.027)	-0.002 (0.123)
Control mean	0.958	5.440	0.968	1.000	0.880	2.569
Observations	625	625	625	625	625	625

Notes: Table A5 shows that attrition levels are very low and mostly balanced across treatment groups. Columns indicate different attrition measures. Columns 1 to 4 test for differential attrition during the treatment period. Columns 5 and 6 test for differential attrition during the post-treatment period. Heteroskedasticity robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Characteristics of employers

	Control	Non-compliers	Compliers	Δ	p(Non-compliers=Compliers)
Firm characteristics					
Age of firm (in years)	7.45	7.14	6.88	-0.260	0.799
# of employees	15.12	13.15	18.01	4.861	0.002
Share of white-collar employees	0.13	0.14	0.16	0.025	0.222
Av. salary of workers	3216.24	2983.93	2935.97	-47.965	0.836
Av. salary of white-collar workers	5183.02	4791.61	5568.57	776.959	0.143
Manufacturing sector	0.52	0.56	0.25	-0.310	0.000
Service sector (retail, hospitality)	0.27	0.25	0.37	0.113	0.049
Health Sector	0.07	0.09	0.26	0.170	0.001
Hiring practices					
Uses formal hiring channels	0.10	0.08	0.09	0.013	0.700
Uses network hiring channels	0.81	0.78	0.82	0.039	0.410
Uses employment agencies	0.36	0.37	0.57	0.207	0.001
Manager expectations					
Expected number of hires in next 3 months	3.06	3.36	4.84	1.483	0.111
Positive bus. outlook next 3 months	0.62	0.59	0.68	0.086	0.136
Positive bus. outlook next 12 months	0.79	0.75	0.82	0.073	0.136
Manager characteristics					
Female	0.30	0.22	0.28	0.056	0.297
Manager age	34.98	35.40	35.83	0.428	0.718
Manager has univ. degree	0.42	0.45	0.56	0.117	0.053
Raven's Matrix score (out of 10)	5.01	4.71	5.34	0.631	0.039

Notes: Table A6 shows that firms who make use of the vacancy posting subsidy work in different sectors and differ on further characteristics. All variables are measured at baseline. Column 5 tests for equality of means between compliers and non-compliers.

Table A7: Impact on other employee search methods

	Network			Agency			Premises		
	(1) Any	(2) # vacs	(3) % vacs	(4) Any	(5) # vacs	(6) % vacs	(7) Any	(8) # vacs	(9) % vacs
Panel A: Pooled Treatments									
Treatment	-0.078** (0.036) [0.070]*	-0.055 (0.063) [0.114]	-0.094* (0.054) [0.079]*	-0.063** (0.032) [0.070]*	-0.182** (0.092) [0.070]*	-0.139*** (0.050) [0.051]*	-0.044** (0.021) [0.070]*	-0.079* (0.047) [0.079]*	-0.066** (0.031) [0.070]*
Control mean	0.269	0.398	0.462	0.199	0.449	0.338	0.083	0.148	0.115
Observations	621	621	288	621	621	288	621	621	288
Panel B: Separated Treatments									
Treatment	-0.027 (0.042) [0.399]	0.046 (0.082) [0.399]	-0.016 (0.063) [0.554]	-0.069* (0.036) [0.152]	-0.237*** (0.090) [0.038]**	-0.166*** (0.054) [0.020]**	-0.040* (0.024) [0.191]	-0.047 (0.058) [0.390]	-0.054 (0.035) [0.204]
Treatment × screening	-0.105*** (0.039) [0.030]**	-0.207*** (0.075) [0.030]**	-0.154** (0.062) [0.034]**	0.011 (0.034) [0.711]	0.111 (0.098) [0.384]	0.051 (0.051) [0.384]	-0.008 (0.019) [0.711]	-0.066 (0.043) [0.227]	-0.024 (0.027) [0.384]
Total treatment effect screening	-0.132*** (0.039) [0.006]***	-0.161** (0.063) [0.021]**	-0.170*** (0.062) [0.021]**	-0.058 (0.037) [0.059]*	-0.126 (0.117) [0.070]*	-0.114* (0.058) [0.030]**	-0.048** (0.023) [0.026]**	-0.113*** (0.043) [0.021]**	-0.078** (0.032) [0.023]**
Control mean	0.269	0.398	0.462	0.199	0.449	0.338	0.083	0.148	0.115
Observations	621	621	288	621	621	288	621	621	288

Notes: Table A7 shows that vacancy posting subsidies led to a substitution away from other search channels. Panel A shows pooled treatment effects. Panel B shows heterogeneity by whether the intervention includes a skill screening component. Columns 1 to 3 show effects on network-based search. Columns 4 to 6 show effects on agency-based search. Columns 7 to 9 show effects on advertising the vacancy on firms' premises. Columns 1, 4, and 7 show effects on the fraction of firms advertising at least one vacancy through the respective channel. Columns 2, 5, and 8 show effects on the number of vacancies advertised through the respective channel. Columns 3, 6, and 9 show effects on the fraction of vacancies advertised through the respective channel. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Effects on characteristics of hires

	(1) Salary (ETB, IHS)	(2) Satisfaction	(3) Share female	(4) Share with degree
Panel A: Pooled				
Treatment	0.035 (0.083) [1.000]	-0.034 (0.126) [1.000]	-0.023 (0.057) [1.000]	-0.007 (0.054) [1.000]
Control mean	8.165	0.020	0.586	0.408
Observations	232	236	250	250

Notes: **Table A8 shows that the intervention had no measurable effect on the characteristics of new hires.** It displays the effect of our intervention on the characteristics of new hires. Column 1 shows effects on the inverse hyperbolic sine of the average salary of new hires. Column 2 shows the effect on a standardized measure of satisfaction with new hires. Column 3 shows the effect on the fraction of women among new hires. Column 4 shows the effect on the fraction of hires with a university degree among new hires. The outcomes are only defined for firms that hired at least one person during the treatment period. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Baseline search inputs by posting channel

	Pooled	Formal	Non-formal
Search cost index (std.)	-0.000	0.881	-0.016
Search duration (days)	15.668	6.900	15.931
Screening hours	3.572	6.350	3.514
Non-screen hours	0.504	1.250	0.488
Total hours	3.889	7.600	3.816
Screening cost (ETB)	240.831	812.353	230.555
Non-screen cost (ETB)	37.239	158.824	35.042
Total cost (ETB)	262.729	920.000	251.375
Vacancies	1153	32	1112

Notes: **Table A9 reports mean search inputs for baseline vacancies, split by posting channel.** Formal vacancies are those advertised through job boards, newspapers, or internet postings; vacancies filled through employment agencies are excluded from the formal/non-formal split. Search duration is measured in days, hour outcomes are measured in hours, and cost outcomes are measured in Ethiopian Birr (ETB). The search cost index is standardized to have mean zero and standard deviation one in the pooled baseline vacancy sample.

Table A10: Effects on managers' beliefs about formal employee search

	Applicant quality			Applicant numbers (standardized)		
	(1) Index	(2) WC	(3) Non-WC	(4) Index	(5) WC	(6) Non-WC
Treatment	-0.169** (0.084) [0.072]*	-0.133 (0.084) [0.072]*	-0.183** (0.084) [0.072]*	-0.214* (0.111) [0.091]*	-0.198* (0.115) [0.091]*	-0.203* (0.110) [0.091]*
Control mean	0.110	0.087	0.120	0.141	0.131	0.134
Observations	605	605	605	561	553	560

Notes: **Table A10 shows that managers negatively update their beliefs about the returns to formal search.** It displays the treatment effects on beliefs about the quality and number of applicants obtained through formal search channels. Columns 1 to 3 show the impacts on beliefs about applicant quality. Applicant quality is measured by binary variables equal to one if managers believe that they can obtain better quality candidates through different formal search channels relative to network-based hiring. Columns 4 to 6 show the impacts on beliefs about absolute applicant numbers. All variables are normalized sums of non-missing normalized beliefs across different formal search channels (online, job board, newspaper) and vacancy type (white-collar, blue collar, pink collar). Number of observations varies for beliefs about applicant numbers due to "don't know" answers. Minimum q-values from two-stage false discovery rate correction within families (columns 1-3 and 4-6) are displayed in brackets. Heteroskedasticity robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Treatment effects on the number of applicants

	(1) # applicants per vacancy	(2) # applicants per vacancy any vacs
Treatment	2.362* (1.340) [0.085]*	5.803** (2.890) [0.085]*
Control mean	2.361	4.766
Observations	621	288

Notes: Table A11 reports treatment effects on the number of applicants per vacancy, measured on the firm-level. Column 1 uses the total number of applicants per vacancy reported by firms in the full sample. Column 2 uses the same outcome conditional on the firm having created at least one vacancy during the treatment period. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Expected and realized earnings

	Applicant data		Realized salary data		
	(1) Reservation wage	(2) Wage expectation	(3) All firms	(4) Large firms	(5) Average salary at bl
All vacancies	5059	5490	.	.	3563
Vacancies with hires	4066	4700	3256	3935	2960
Vacancies without hires	5601	5907	.	.	3665
<u>Δ: without hires - with hires</u>					
Raw	1535	1207			
Adjusted	1413	976			

Notes: Table A12 shows that applicants are generally overoptimistic about offered salaries. It compares average reservation salaries and salary expectations to realized salaries and average baseline salaries. All values are in Ethiopian Birr (ETB) per month (100 ETB were worth around 5.21 USD PPP in 2019). Samples are restricted to the vacancy subsidy plus screening treatment group because reservation salary and salary expectation data are only available for applicants applying to vacancies posted in the screening group. Columns 1 and 2 are applicant-level averages (applicants to vacancies with hires but without salary information are excluded to make results comparable to column 3). Column 3 is the average salary of newly hired employees for vacancies posted during the treatment period (at the vacancy level). Column 4 is the average salary of newly hired employees for vacancies posted during the treatment period by the firms with an above-median number of employees. The sample in column 5 displays firm-level baseline wages with the sample defined to be comparable to columns 1 to 3. Δ describes the differences in applicant expectations between unfilled and filled vacancies. The adjusted difference controls for vacancy type (white-collar, blue-collar, or pink-collar) and for popularity (a dummy for receiving an above median number of applications).

Table A13: Treatment effects by hiring professionalism

	Vacancy creation						Hires				Fraction filled		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
	Any	#	Any wc	# wc	Any non-wc	# non-wc	Any	#	Any wc	# wc	Any non-wc	# non-wc	pooled
Panel A. Above-median split													
Treatment below-median professionalism	-0.083 (0.059)	0.052 (0.194)	0.058** (0.027)	0.104** (0.051)	-0.115* (0.059)	-0.052 (0.181)	-0.091 (0.059)	-0.140 (0.215)	0.018 (0.024)	0.064 (0.059)	-0.103* (0.058)	-0.204 (0.200)	-0.134** (0.056)
Treatment × above-median professionalism	0.064 (0.084)	0.136 (0.343)	0.027 (0.051)	0.135 (0.131)	0.087 (0.084)	-0.005 (0.297)	0.020 (0.084)	-0.150 (0.345)	0.002 (0.045)	-0.121 (0.125)	0.028 (0.082)	-0.029 (0.309)	-0.127 (0.081)
Above-median professionalism treatment effect	-0.019 (0.060)	0.188 (0.283)	0.085** (0.043)	0.239** (0.120)	-0.028 (0.059)	-0.057 (0.235)	-0.071 (0.060)	-0.290 (0.269)	0.020 (0.038)	-0.057 (0.109)	-0.074 (0.058)	-0.233 (0.236)	-0.260*** (0.058)
Panel B. Continuous index													
Treatment (at index = 0)	-0.051 (0.042)	0.121 (0.172)	0.070*** (0.026)	0.168** (0.065)	-0.070* (0.042)	-0.049 (0.148)	-0.080* (0.042)	-0.210 (0.173)	0.018 (0.022)	0.000 (0.063)	-0.086** (0.041)	-0.210 (0.154)	-0.199*** (0.041)
Treatment × index	0.019 (0.042)	0.081 (0.160)	0.014 (0.026)	0.043 (0.067)	0.030 (0.040)	0.034 (0.134)	0.022 (0.040)	0.009 (0.163)	0.016 (0.023)	-0.054 (0.073)	0.027 (0.038)	0.062 (0.136)	-0.026 (0.047)
Treatment at +1 s.d. professionalism	-0.032 (0.060)	0.203 (0.258)	0.084* (0.044)	0.211* (0.121)	-0.040 (0.057)	-0.015 (0.200)	-0.058 (0.058)	-0.202 (0.248)	0.034 (0.039)	-0.054 (0.126)	-0.059 (0.054)	-0.148 (0.193)	-0.225*** (0.070)
Control mean	0.498 618	1.158 618	0.079 618	0.144 618	0.451 618	1.014 618	0.456 618	1.223 618	0.070 618	0.153 618	0.414 618	1.070 618	0.877 287
Observations													877 251

Notes: Table A13 reports treatment heterogeneity by the pre-treatment hiring professionalism index. Panel A reports heterogeneity by an above-median split of the original two-component index combining the hiring-process score and the structured-selection measure. Treatment is the effect for below-median firms; the interaction term reports the differential effect for above-median firms; the treatment effect for above-median firms is reported in the lower part of Panel A. Panel B reports the analogous specification using the continuous index. The treatment coefficient is the treatment effect at index value zero, and the lower part of Panel B reports the implied treatment effect at one standard deviation above the mean. The full baseline sample contains $N = 625$ firms. Smaller samples arise from variable-specific missingness, and filling-rate outcomes are mechanically restricted to firms with vacancies. Heteroskedasticity robust standard errors are displayed in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Heterogeneous treatment effects on vacancy creation and hires.

	Firm characteristics			Sector		Hiring practices			Expectations			
	(1) Bus. age	(2) # bl.employees	(3) % wc employees	(4) Manufacturing	(5) Service	(6) Exp. hires (3m)	(7) Pos. bus. outlook (3m)	(8) Pos. bus. outlook (12m)	(9) Female	(10) Age	(11) Univ. degree	(12) Raven's score
Panel A: Impact on number of vacancies posted												
Treatment	0.008 (0.226)	-0.272 (0.287)	0.190 (0.217)	0.250 (0.287)	0.124 (0.192)	0.018 (0.216)	0.486** (0.228)	0.545** (0.275)	0.040 (0.220)	0.493 (0.753)	0.106 (0.206)	-0.366 (0.453)
Treatment × hetero. var	0.018 (0.026)	0.029 (0.024)	-0.517 (1.185)	-0.291 (0.340)	-0.033 (0.392)	0.020 (0.053)	-0.509 (0.327)	-0.469 (0.349)	0.242 (0.312)	-0.011 (0.022)	0.011 (0.357)	0.055 (0.043)
Hetero. var	0.006 (0.015)	0.034* (0.019)	0.672 (1.021)	-0.673*** (0.260)	0.828*** (0.292)	0.045 (0.031)	0.777*** (0.221)	0.825*** (0.230)	-0.395* (0.226)	0.014 (0.019)	0.249 (0.277)	-0.045 (0.035)
Control mean	1.153 613	1.153 609	1.153 621	1.153 621	1.153 621	1.153 620	1.153 607	1.153 580	1.153 621	1.153 621	1.153 621	1.153 621
Panel B: Impact on number of hires												
Treatment	-0.229 (0.230)	-0.640** (0.265)	-0.106 (0.243)	-0.348 (0.262)	-0.129 (0.193)	-0.274 (0.200)	0.183 (0.206)	0.225 (0.277)	-0.243 (0.210)	0.221 (0.661)	-0.203 (0.196)	-0.535 (0.389)
Treatment × hetero. var	0.004 (0.024)	0.031 (0.022)	-0.730 (1.148)	0.242 (0.342)	-0.325 (0.393)	0.011 (0.048)	-0.541* (0.319)	-0.491 (0.351)	0.064 (0.350)	-0.012 (0.018)	-0.045 (0.360)	0.036 (0.041)
Hetero. var	0.003 (0.019)	0.032** (0.016)	0.135 (1.056)	-0.642** (0.280)	0.928*** (0.318)	0.039 (0.036)	0.773*** (0.249)	0.761*** (0.279)	-0.267 (0.288)	0.003 (0.015)	0.289 (0.301)	-0.014 (0.033)
Control mean	1.218 613	1.218 609	1.218 621	1.218 621	1.218 621	1.218 620	1.218 607	1.218 580	1.218 621	1.218 621	1.218 621	1.218 621

Notes: Table A14 shows that there are no further heterogeneous treatment effects on the number of posted vacancies and on the number of hires by baseline firm and manager characteristics. The dependent variable in Panel A is the number of created vacancies. The dependent variable in Panel B is the number of hires. Columns indicate heterogeneity variables. Columns 1 to 3 show heterogeneity by firm characteristics (firm age, size, and share of white-collar employees). Columns 4 and 5 show heterogeneity by sector of the firm (manufacturing and service). Columns 6 to 8 show effects by managers' baseline expectations (expected number of hires and dummies indicating a positive business outlook for the next 3 and 12 months). Columns 9 to 12 show heterogeneity by manager characteristics (gender, age, education, and Raven's score). Heteroskedasticity robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Treatment effects on expectations by baseline use of formal channels

	Expected applicant quality (index)			Expected applicant numbers (index)		
	(1) Pooled	(2) wc	(3) non-wc	(4) Pooled	(5) wc	(6) non-wc
Treatment	-0.168* (0.089)	-0.137 (0.090)	-0.178** (0.089)	-0.158 (0.108)	-0.150 (0.107)	-0.155 (0.112)
Treatment × Used formal channels at baseline	0.137 (0.251)	0.159 (0.251)	0.097 (0.247)	-0.465 (0.498)	-0.383 (0.565)	-0.393 (0.419)
Used formal channels at baseline	0.362* (0.198)	0.260 (0.200)	0.418** (0.194)	0.572 (0.480)	0.510 (0.551)	0.483 (0.399)
Treatment effect: Used formal channels at baseline	-0.032 (0.235)	0.021 (0.234)	-0.081 (0.231)	-0.622 (0.486)	-0.533 (0.555)	-0.548 (0.404)
Control mean	0.109	0.086	0.118	0.141	0.131	0.134
Observations	606	606	606	561	553	560

Notes: Table A15 shows that effects on beliefs about applicant quality are driven by firms without experience of using formal search channels. It displays the effects on standardized beliefs about the quality of applicants obtained through formal search channels (columns 1 to 3) and the number of applicants obtained (columns 4 to 6) parallel to our main outcomes. At baseline, 8.8% of firms used formal search channels. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Comparing experienced and inexperienced firms

	Inexperienced	Experienced	Δ	p(Inexperienced=Experienced)
Firm characteristics				
Age of firm (in years)	6.76	11.66	4.904	0.008
# of employees	13.69	22.84	9.149	0.000
Share of white-collar employees	0.13	0.23	0.096	0.000
Manufacturing sector	0.51	0.45	-0.058	0.413
Service sector (retail, hospitality)	0.28	0.24	-0.041	0.499
Health Sector	0.11	0.11	0.000	0.994
Average salary (1000s Birr)	3.37	4.18	0.815	0.068
Hiring practices				
Uses formal hiring channels	0.00	1.00	1.000	.
Uses network hiring channels	0.80	0.71	-0.093	0.145
Uses other hiring channels	0.67	0.65	-0.019	0.776
Manager expectations				
Expected number of hires in next 3 months	3.35	4.60	1.250	0.183
Positive bus. outlook next 3 months	0.62	0.60	-0.017	0.807
Positive bus. outlook next 12 months	0.77	0.78	0.009	0.883
Manager characteristics				
Female	0.25	0.29	0.038	0.550
Manager age	34.96	38.96	3.999	0.018
Manager has univ. degree	0.42	0.82	0.399	0.000
Raven's Matrix score (out of 10)	4.85	5.35	0.493	0.194

Notes: Table A16 shows that firms with experience in using formal hiring channels are larger and pay higher wages. It compares variable means across firms who regularly used formal search channels at baseline to those who did not. p-values are based on t-tests with heteroskedasticity robust standard errors.

Table A17: Skill screening and formalization of employee search

	Take-up			Formal search			Other search		
	(1) Any	(2) Any any vacs	(3) # vacs	(4) Any	(5) # vacs	(6) % vacs	(7) Any	(8) # vacs	(9) % vacs
Treatment	0.203*** (0.028) [0.001]***	0.472*** (0.053) [0.001]***	0.522*** (0.107) [0.001]***	0.152*** (0.032) [0.001]***	0.446*** (0.135) [0.001]***	0.297*** (0.050) [0.001]***	-0.129*** (0.048) [0.002]***	-0.278** (0.140) [0.009]***	-0.275*** (0.046) [0.001]***
Treatment × screening	0.024 (0.041) [1.000]	0.017 (0.075) [1.000]	0.069 (0.161) [1.000]	0.034 (0.041) [1.000]	0.027 (0.167) [1.000]	0.025 (0.064) [1.000]	-0.034 (0.047) [1.000]	-0.187 (0.132) [1.000]	-0.112* (0.063) [1.000]
Total treatment effect screening	0.227*** (0.030) [0.001]***	0.489*** (0.052) [0.001]***	0.591*** (0.121) [0.001]***	0.186*** (0.034) [0.001]***	0.473*** (0.144) [0.001]***	0.323*** (0.050) [0.001]***	-0.163*** (0.048) [0.001]***	-0.465*** (0.140) [0.001]***	-0.387*** (0.049) [0.001]***
Control mean	0.000	0.000	0.000	0.051	0.144	0.070	0.486	1.046	0.951
Observations	621	288	621	621	621	288	621	621	288

Notes: Table A17 shows that skill screening does not significantly affect the formalization of employee search. Column 1 shows the fraction of firms posting at least one vacancy through our intervention. Column 2 shows the number of vacancies posted through our intervention conditional on using the subsidy for at least one vacancy. Column 3 shows the number of vacancies for which the vacancy subsidy was used. Columns 4 to 6 show the impact of the vacancy subsidy on formal employee search. Columns 7 to 9 show the impact of the vacancy subsidy on other methods of employee search (exclusively network-based). Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Skill screening and vacancy creation

	Pooled			White collar			Non-white collar	
	(1) Any	(2) # vacs	(3) # vacs any	(4) Any vac	(5) # vacs	(6) % vacs	(7) Any vac	(8) # vacs
Treatment	-0.065 (0.049) [0.097]*	0.142 (0.197) [0.253]	0.684** (0.323) [0.071]*	0.071** (0.031) [0.071]*	0.214** (0.095) [0.071]*	0.132** (0.051) [0.071]*	-0.092* (0.048) [0.071]*	-0.072 (0.161) [0.311]
Treatment × screening	0.035 (0.050) [1.000]	-0.037 (0.225) [1.000]	-0.305 (0.390) [1.000]	0.002 (0.036) [1.000]	-0.085 (0.103) [1.000]	-0.028 (0.056) [1.000]	0.047 (0.048) [1.000]	0.043 (0.193) [1.000]
Total treatment effect screening	-0.031 (0.049) [0.798]	0.105 (0.212) [0.799]	0.379 (0.363) [0.551]	0.073** (0.031) [0.135]	0.129* (0.069) [0.137]	0.104** (0.048) [0.135]	-0.045 (0.049) [0.551]	-0.029 (0.190) [1.000]
Control mean	0.495	1.153	2.327	0.079	0.144	0.119	0.449	1.009
Observations	621	621	288	621	621	288	621	621

Notes: Table A18 shows that the screening add-on did not significantly affect vacancy creation. Columns 1 to 3 show effects on vacancy creation by firms. Column 1 shows effects on a dummy indicating at least one created vacancy. Column 2 shows the effect on the number of created vacancies. Column 3 shows effects on the number of vacancies conditional on creating at least one vacancy. Columns 4 to 6 show the impact on the creation of white-collar vacancies. Column 4 shows effects on a dummy indicating at least one created white-collar vacancy. Column 5 shows the effect on the number of created white-collar vacancies. Column 6 shows effects on the fraction of white-collar vacancies. Columns 7 and 8 show the impact on the creation of non-white-collar vacancies. Column 7 shows effects on a dummy indicating at least one created non-white-collar vacancy. Column 8 shows the effect on the number of created non-white-collar vacancies. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Skill screening and hiring

	Pooled			White collar				Non-white collar		
	(1) Any hire	(2) # hires	(3) % vacs filled	(4) Any hire	(5) # hires	(6) % hires	(7) % vacs filled	(8) Any hire	(9) # hires	(10) % vacs filled
Treatment	-0.082* (0.048) [0.118]	-0.140 (0.200) [0.273]	-0.178*** (0.049) [0.003]***	0.027 (0.027) [0.220]	0.031 (0.075) [0.373]	0.085 (0.053) [0.121]	-0.321*** (0.119) [0.025]**	-0.098** (0.047) [0.066]*	-0.171 (0.180) [0.220]	-0.139*** (0.051) [0.025]**
Treatment × screening	0.007 (0.048) [1.000]	-0.143 (0.197) [1.000]	-0.049 (0.058) [1.000]	-0.016 (0.028) [1.000]	-0.052 (0.067) [1.000]	-0.046 (0.057) [1.000]	-0.074 (0.119) [1.000]	0.024 (0.047) [1.000]	-0.091 (0.178) [1.000]	-0.054 (0.061) [1.000]
Total treatment effect screening	-0.075 (0.048) [0.172]	-0.283 (0.195) [0.172]	-0.227*** (0.051) [0.001]***	0.011 (0.026) [0.283]	-0.021 (0.066) [0.287]	0.039 (0.049) [0.265]	-0.395*** (0.119) [0.003]***	-0.074 (0.048) [0.172]	-0.262 (0.175) [0.172]	-0.193*** (0.054) [0.002]***
Control mean	0.454	1.218	0.877	0.069	0.153	0.118	0.847	0.412	1.065	0.877
Observations	621	621	288	621	621	250	78	621	621	252

Notes: Table A19 shows that skill screening did not significantly affect hiring outcomes. Columns 1 to 3 show effects on average hiring outcomes. Column 1 shows effects on a dummy indicating at least one hire during the treatment period. Column 2 shows the effect on the number of hires. Column 3 shows effects on the fraction of filled vacancies. Columns 4 to 7 show the impact on white-collar hiring outcomes. Column 4 shows effects on a dummy indicating at least one white-collar hire. Column 5 shows the effect on the number of white-collar hires. Column 6 shows effects on the fraction of white-collar hires. Column 7 shows effects on the fraction of filled white-collar vacancies. Columns 8 to 10 show the impact on non-white-collar hires. Column 8 shows effects on a dummy indicating at least one non-white-collar hire. Column 9 shows the effect on the number of non-white-collar hires. Column 10 shows effects on the fraction of filled non-white-collar vacancies. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A20: Comparing treated postings and formal control vacancies

	Treated non-template	Treated template	Control formal	Δ T-NT	Δ C-T	Δ C-NT	P All equal
Harmonized posting characteristics							
Requires specific qualification	1.000	1.000	0.636	0.000	-0.364	-0.364	0.000
Additional requirements: language	0.000	0.105		0.105			0.144
Additional requirements: location	0.100	0.211		0.111			0.352
Additional requirements: equipment	0.000	0.000		0.000			
Additional requirements: other	0.500	0.579		0.079			0.632
Requires more than 3 years of experience	0.150	0.000	0.773	-0.150	0.773	0.623	0.000
Senior job title	0.150	0.158	0.318	0.008	0.160	0.168	0.333
Informational content / richness proxies							
Includes descriptive job details	0.400	0.526	0.409	0.126	-0.117	0.009	0.686
Includes salary / compensation info	0.450	0.000	0.409	-0.450	0.409	-0.041	0.002
States compensation numerically	0.050	0.000	0.364	-0.050	0.364	0.314	0.001
Includes number required / quantity	0.200	0.947		0.747			0.000
Includes application instructions	0.300	0.842		0.542			0.000
Includes address / location details	1.000	0.632		-0.368			0.002
Includes employment type / contract terms	0.150	0.263		0.113			0.395
Posting text length (100 chars)	5.076	6.324		1.248			0.041
Structured content lines	12.800	5.895	2.273	-6.905	-3.622	-10.527	0.000

Notes: Table A20 compares three groups of postings: treated non-template postings, treated template postings, and formal postings observed among control-group firms. The top panel reports applicant requirements and vacancy characteristics. The bottom panel reports applicant-facing information contained in the posting itself. Where a variable is not recoverable for the control postings from vacancy records, the corresponding control mean is left blank. The final column reports the p-value from a joint test of equality across the three group means.

Table A21: Treatment effects on vacancy creation by creation reason

	Replace leaving worker			Grow business			Spread workload		
	(1) pooled	(2) wc	(3) non-wc	(4) pooled	(5) wc	(6) non-wc	(7) pooled	(8) wc	(9) non-wc
Treatment	-0.170 (0.117)	0.052* (0.030)	-0.224** (0.107)	-0.074 (0.057)	-0.022 (0.024)	-0.053 (0.045)	0.022 (0.042)	0.002 (0.015)	0.020 (0.040)
Control mean	0.745	0.069	0.676	0.218	0.046	0.171	0.139	0.023	0.116
Observations	621	621	621	621	621	621	621	621	621

Notes: Table A21 shows that the increase in white-collar vacancies is driven by vacancies to replace workers who left. Columns 1 to 3 show effects on the number of vacancies created to replace a worker who left the company. Columns 4 to 6 show effects on the number of vacancies created to expand the business. Columns 7 to 9 show effects on the number of vacancies created to spread the existing workload. Columns 2, 5, and 8 show the impact on the number of respective white-collar vacancies. Columns 3, 6, and 9 show the impact on the number of non-white-collar vacancies. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A22: Effects on candidate search inputs

	Index	Days	Hours			Cost (ETB)		
	(1) Search costs	(2) Search duration	(3) Screening	(4) Non-screening	(5) Total	(6) Screening	(7) Non-screening	(8) Total
Treatment	-0.019 (0.144) [0.897]	-1.055 (1.170) [0.736]	-1.051 (0.860) [0.717]	1.277 (0.843) [0.717]	0.168 (1.295) [0.897]	22.323 (118.531) [0.897]	19.060 (17.208) [0.717]	48.872 (120.154) [0.897]
Control mean	0.000	4.951	4.410	1.046	5.448	228.528	17.512	241.887
Observations	240	234	227	226	227	236	233	234

Notes: Table A22 shows that the intervention did not affect firms’ search inputs or search duration conditional on creating at least one vacancy. It displays the effect of our intervention on candidate search inputs conditional on creating at least one vacancy. Column 1 shows impacts on a variance-covariance weighted index of search cost with the four components in columns 3, 4, 6, and 7. Column 2 shows the impact on search duration in days conditional on hiring. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Effects on average monthly salaries

	Average salaries at endline (ihs)		
	(1) Pooled	(2) White collar	(3) Non-white collar
Treatment	0.120* (0.063) [0.094]*	-0.015 (0.070) [0.381]	0.121* (0.062) [0.094]*
Control mean	8.412	8.944	8.327
Observations	597	418	596

Notes: Table A23 shows that firms in the treatment group have a slightly higher average salary level at endline. It displays the effect of our intervention on average monthly salaries at endline (transformed using the inverse hyperbolic sine). Column 2 shows the impact on white-collar wages conditional on having white-collar employees. Column 3 shows the impact on non-white-collar wages conditional on having non-white-collar employees. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. Heteroskedasticity robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A24: Effects on employee turnover

	Employees left		Leaving reasons		
	(1)	(2)	(3)	(4)	(5)
	Any	#	Personal	Better opportunities	Fired for performance
Treatment	-0.002 (0.041) [0.920]	-0.361 (0.292) [0.767]	-0.075** (0.035) [0.096]*	-0.012 (0.022) [0.643]	-0.018 (0.019) [0.513]
Control mean	0.597	2.435	0.241	0.079	0.060
Observations	621	621	621	621	621

Notes: Table A24 displays the effect of our intervention on employee turnover. Column 1 shows the impact on a dummy variable indicating any turnover during this period. Column 2 shows the impact on the number of employees who left the firm (winsorized at the 99th percentile). Columns 3 to 5 show the impact of mentioning the respective reasons as important for employee turnover at least once during phone or endline surveys. Column 3 displays the impact on mentioning personal reasons, column 4 displays the impact on mentioning leaving for better opportunities, and column 5 shows the impact on mentioning firing workers for bad performance. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction within families (columns 1 and 2 and 3 to 5) are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A25: Effects on employee numbers and shares

	(1) # of employees	(2) Share of WC employees
Treatment	-2.488 (1.605) [0.121]	0.024** (0.011) [0.068]*
Control mean	16.818	0.099
Observations	606	600

Notes: Table A25 shows that treated firms have a higher share of white-collar employees at **endline**. It displays the effect of our intervention on the number of employees and the share of white-collar employees. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A26: Impact on post-treatment behavior by baseline usage of formal search

	Vacancy posting					Hires		
	(1) Any	(2) #	(3) % formal	(4) % network	(5) % wc	(6) Any	(7) #	(8) % wc
Treatment	-0.088** (0.037)	-0.188*** (0.070)	0.002 (0.033)	0.022 (0.100)	0.197*** (0.070)	-0.091** (0.037)	-0.274*** (0.088)	0.177** (0.069)
Treatment × Used formal channels at baseline	0.147 (0.125)	0.153 (0.274)	-0.288 (0.293)	-0.213 (0.268)	-0.043 (0.282)	0.115 (0.122)	0.238 (0.279)	-0.121 (0.289)
Used formal channels at baseline	-0.034 (0.092)	0.056 (0.218)	0.469* (0.256)	0.024 (0.221)	0.364 (0.223)	-0.034 (0.092)	-0.033 (0.225)	0.364 (0.224)
Treatment effect: Used formal channels at baseline	0.060 (0.119)	-0.036 (0.265)	-0.286 (0.292)	-0.190 (0.249)	0.155 (0.274)	0.024 (0.116)	-0.036 (0.265)	0.056 (0.280)
Control mean	0.221	0.379	0.075	0.312	0.087	0.221	0.458	0.087
Observations	554	554	95	95	95	554	554	93

Notes: Table A26 shows heterogeneous treatment effects on post-treatment behavior by baseline usage of formal search channels. It displays the effects of the vacancy subsidy intervention on post-treatment vacancy posting and hiring in the two months following the four-month treatment period. “Used formal channels at baseline” is a dummy indicating the use of formal search channels. Columns 1 to 5 show treatment effects on vacancy creation. Column 1 shows the effects on creating any vacancy post-treatment. Column 2 shows the effect on the number of created vacancies. Column 3 shows the effect on the fraction of formally posted vacancies. Column 4 shows the effect on the fraction of network vacancies. Column 5 shows the effect on the fraction of white-collar vacancies. Columns 6 to 8 show effects on hiring outcomes. Column 6 shows the effect on having any hire post-treatment. Column 7 shows the effect on the number of post-treatment hires. Column 8 shows the effect on the fraction of white-collar hires. Outcomes in columns 3, 4, 5 and 8 are only defined for firms with at least one created vacancy or hire. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A27: Dynamics of the Effects on Vacancy Creation and Hiring Levels

	# vacs			# hires	
	(1) take-up	(2) all vacs	(3) wc vacs	(4) all hires	(5) wc hires
Treatment	0.109*** (0.026)	-0.127 (0.080)	-0.007 (0.031)	-0.081 (0.089)	-0.018 (0.028)
Treatment $\times$ Late period	-0.049* (0.028)	-0.009 (0.098)	0.029 (0.040)	-0.119 (0.131)	0.008 (0.051)
Late period	0.000 -	0.162** (0.076)	-0.014 (0.030)	0.245** (0.102)	0.014 (0.042)
Treatment effect during late period	0.059*** (0.013)	-0.136 (0.098)	0.022 (0.032)	-0.201* (0.121)	-0.009 (0.048)
Control mean	0.000	0.576	0.072	0.609	0.076
Observations	1242	1242	1242	1242	1242

Notes: **Table A27 shows that take-up but not treatment effects decline over the treatment period.** Later period captures firm activity after the first two months of treatment. Data include two observations per firm: one capturing vacancy creation and hiring during the first two months of the treatment period and one capturing activity thereafter. Standard errors clustered at the firm level are displayed in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A28: Dynamics of the Effects on Vacancy Creation and Hiring Percentages

	% vacancies			% hires
	(1) take-up	(2) wc vacs	(3) filled	(4) wc hires
Treatment	0.256*** (0.043)	0.099* (0.057)	-0.052 (0.052)	0.051 (0.059)
Treatment $\times$ Late period	-0.119** (0.049)	-0.025 (0.070)	-0.021 (0.064)	-0.002 (0.073)
Late period	0.000 -	-0.030 (0.047)	-0.036 (0.037)	-0.045 (0.049)
Treatment effect during late period	0.137*** (0.030)	0.075 (0.045)	-0.073 (0.055)	0.048 (0.046)
Control mean	0.000	0.106	0.859	0.109
Observations	345	345	345	297

Notes: **Table A28 shows that take-up but not treatment effects decline over the treatment period.** Later period captures firm activity after the first two months of treatment. Data include two observations per firm: one capturing vacancy creation and hiring during the first two months of the treatment period and one capturing activity thereafter. Outcomes are conditional on creating at least one vacancy (columns 1 to 3) and conditional on hiring at least one employee (column 4). Standard errors clustered at the firm level are displayed in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A29: Effects on willingness to pay for services

	(1) Subsidy	(2) Formal posting
Treatment	69.321 (61.784) [0.972]	0.053 (0.078) [0.972]
Control mean	278.565	-0.035
Observations	604	594

Notes: **Table A29 shows that the intervention had no significant effect on firms' willingness-to-pay for formal vacancy postings.** It displays the effect of our intervention on firms' incentivized willingness-to-pay for an additional three months of the subsidy treatment in column 1. Column 2 displays treatment effects on a hypothetical willingness-to-pay for formal vacancy posting (calculated as the standardized average willingness-to-pay for advertising on offline and online job boards and in the major newspaper "The Reporter"). Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A30: Effects on downstream firm outcomes

	(1) Profit (IHS)	(2) Revenue (IHS)	(3) Outlook 3m	(4) Outlook 12m
Treatment	-0.287 (0.229) [1.000]	-0.025 (0.217) [1.000]	-0.021 (0.087) [1.000]	-0.036 (0.089) [1.000]
Control mean	4.128	5.563	0.000	-0.000
Observations	581	580	619	552

Notes: Table A30 shows that the intervention had no significant effect on firms' business outcomes and outlook at endline. It displays the effect of our intervention on the inverse hyperbolic sine of last month's profit (column 1) and revenue (column 2), and standardized business outlook for the next three and twelve months (columns 3 and 4). Profits and revenues are inverse hyperbolic sine (IHS)-transformed. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Results including control variables

This section presents all main results with control variables following the pre-analysis plan. We select control variables separately for each outcome using a LASSO algorithm (Belloni et al., 2013) with the same list of potential control variables. The results remain qualitatively unchanged. The list of control variables includes the following categories:

- Manager characteristics (age, gender, education, ethnicity, general intelligence, beliefs)
- Firm characteristics (firm age, sector, labor demand, management practices, hiring practices)
- Employee characteristics (demographics, collar shares, salaries, turnover)

Table A31: Main impacts on vacancy creation - including control variables

	(1) Any	(2) # vacs	(3) # vacs any
Treatment	-0.046 (0.040) [0.616]	0.082 (0.157) [0.616]	0.384 (0.259) [0.616]
Control mean	0.495	1.153	2.327
Observations	621	621	288

Notes: Table A31 displays the treatment effects on vacancy creation with control variables. Control variables are selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A32: Main impacts on hiring - including control variables

	(1) Any	(2) # hires	(3) % vacs filled
Treatment	-0.079* (0.041) [0.055]*	-0.182 (0.165) [0.098]*	-0.197*** (0.038) [0.001]***
Control mean	0.454	1.218	0.877
Observations	621	621	288

Notes: Table A32 displays the treatment effects on hiring outcomes including control variables. Control variables are selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A33: Composition of vacancy creation - including control variables

	White collar			Non-white collar	
	(1) Any vac	(2) # vacs	(3) % vacs	(4) Any vac	(5) # vacs
Treatment	0.061** (0.024) [0.035]**	0.155** (0.063) [0.035]**	0.045 (0.035) [0.135]	-0.071* (0.039) [0.076]*	-0.054 (0.136) [0.321]
Control mean	0.079	0.144	0.119	0.449	1.009
Observations	621	621	288	621	621

Notes: **Table A33 displays the effect of our intervention on the skill composition of vacancy creation including control variables.** Control variables are selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Columns 1 to 3 show the impact on white-collar vacancies. Columns 4 and 5 show the impact on non-white-collar vacancies. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A34: Composition of new hires - including control variables

	White collar				Non-white collar		
	(1) Any hire	(2) # hires	(3) % hires	(4) % vacs filled	(5) Any hire	(6) # hires	(7) % vacs filled
Treatment	0.014 (0.021) [0.424]	0.011 (0.060) [0.583]	-0.007 (0.036) [0.583]	-0.357*** (0.102) [0.002]***	-0.085** (0.039) [0.054]*	-0.203 (0.146) [0.195]	-0.163*** (0.040) [0.001]***
Control mean	0.069	0.153	0.118	0.847	0.412	1.065	0.877
Observations	621	621	250	78	621	621	252

Notes: Table A34 displays the effect of our intervention on the skill composition of new hires including control variables. All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms. Columns 1 to 4 show the impact on white-collar hires. Columns 5 to 7 show the impact on non-white-collar hires. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A35: Effects on managers' beliefs - including control variables

	Applicant quality			Applicant numbers (standardized)		
	(1) Index	(2) WC	(3) Non-WC	(4) Index	(5) WC	(6) Non-WC
Treatment	-0.156* (0.082) [0.086]*	-0.117 (0.082) [0.154]	-0.201** (0.082) [0.086]*	-0.225** (0.111) [0.086]*	-0.204* (0.122) [0.112]	-0.211* (0.109) [0.086]*
Control mean	0.109	0.086	0.118	0.141	0.131	0.134
Observations	606	606	606	561	553	560

Notes: Table A35 displays the treatment effects on beliefs about the quality and number of applicants obtained through formal search channels including control variables. All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms. Columns 1 to 3 show the impacts on beliefs about applicant quality. Applicant quality is measured by binary variables equal to one if managers believe that they can obtain better quality candidates through different formal search channels relative to network-based hiring. Columns 4 to 6 show the impacts on beliefs about absolute applicant numbers. All variables are normalized sums of non-missing normalized beliefs across different formal search channels (online, job-board, newspaper) and vacancy type (white-collar, blue collar, pink collar). Number of observations varies for beliefs about applicant numbers due to "don't know" answers. White-collar vacancies are defined as "professional, managerial, or administrative" workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A36: Effects on turnover - including control variables

	Employees left		Leaving reasons		
	(1)	(2)	(3)	(4)	(5)
	Any	#	Personal	Better opportunities	Fired for performance
Treatment	-0.005 (0.040) [0.904]	-0.324 (0.277) [0.605]	-0.080** (0.033) [0.077]*	-0.010 (0.022) [0.815]	-0.014 (0.019) [0.767]
Control mean	0.597	2.435	0.241	0.079	0.060
Observations	621	621	621	621	621

Notes: Table A36 displays the effects of the vacancy subsidy intervention on employee turnover including control variables. All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Column 1 shows the impact on a dummy variable indicating any turnover during this period. Column 2 shows the impact on the number of employees who left the firm (winsorized at the 99th percentile). Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A37: Effects on candidate search inputs - including control variables

	Index	Days	Hours			Cost (ETB)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Search costs	Search duration	Screening	Non-screening	Total	Screening	Non-screening	Total
Treatment	-0.019 (0.144) [0.897]	-1.055 (1.170) [0.736]	-1.051 (0.860) [0.717]	1.277 (0.843) [0.717]	0.168 (1.295) [0.897]	22.323 (118.531) [0.897]	19.060 (17.208) [0.717]	48.872 (120.154) [0.897]
Control mean	0.000	4.951	4.410	1.046	5.448	228.528	17.512	241.887
Observations	240	234	227	226	227	236	233	234

Notes: Table A37 displays the effect of our intervention on candidate search inputs including control variables. Table A37 shows that the intervention did not affect firms' search inputs or search duration conditional on creating at least one vacancy. It displays the effect of our intervention on candidate search inputs conditional on creating at least one vacancy. Column 1 shows impacts on a variance-covariance weighted index of search cost with the four components in columns 3, 4, 6, and 7. Column 2 shows the impact on search duration in days conditional on hiring. All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A38: Effects on characteristics of hires - including control variables

	(1) Salary (ETB)	(2) Satisfaction	(3) Share female
Treatment	-155.282 (196.870) [0.646]	-0.034 (0.126) [0.786]	-0.046 (0.047) [0.646]
Control mean	2199.932	0.020	0.586
Observations	232	236	250

Notes: **Table A38 displays the effect of our intervention on the characteristics of new hires including control variables.** All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. The outcomes are only defined for firms that hired at least one person during the treatment period. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

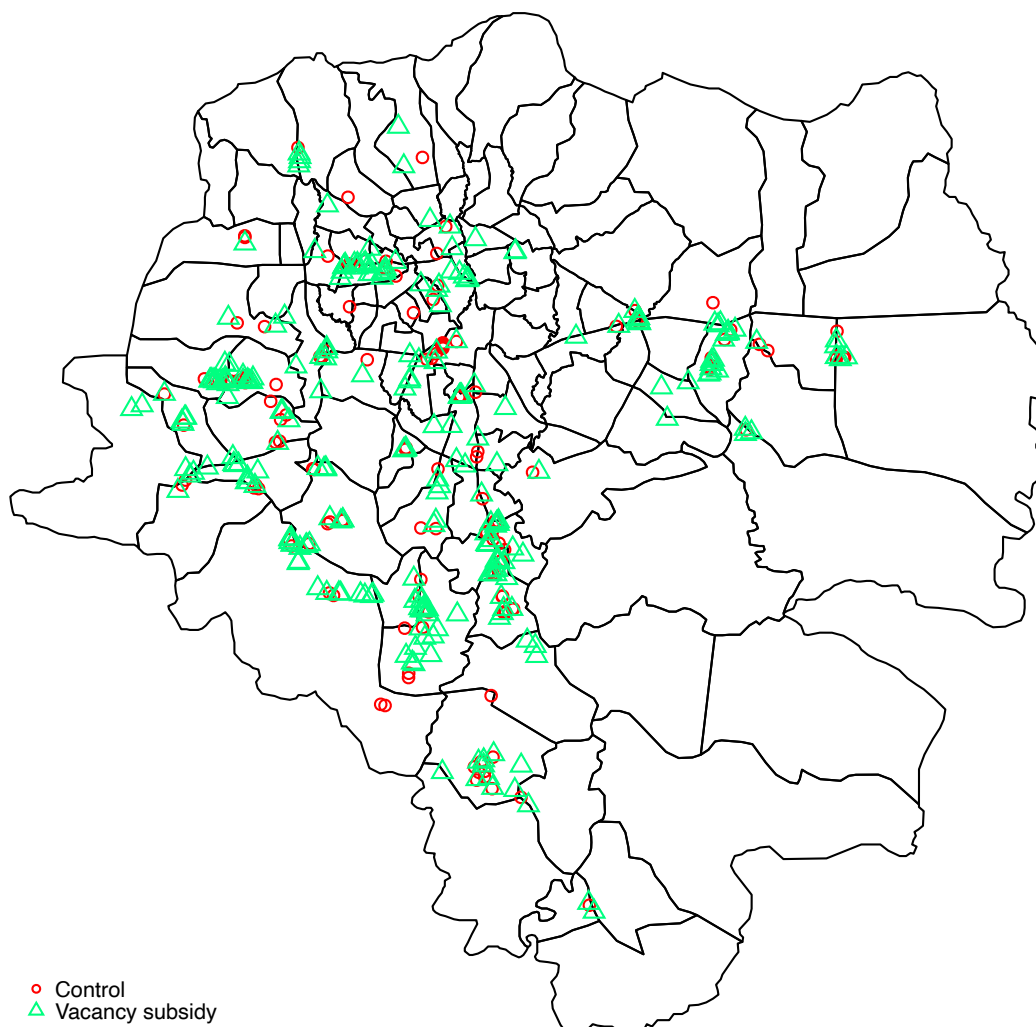
Table A39: Impact on post-treatment behavior - including control variables

	Vacancy posting					Hires		
	(1) Any	(2) #	(3) % formal	(4) % network	(5) % wc	(6) Any	(7) #	(8) % wc
Treatment	-0.074** (0.035) [0.034]**	-0.164** (0.066) [0.025]**	-0.019 (0.048) [0.229]	-0.024 (0.074) [0.229]	0.183*** (0.062) [0.015]**	-0.082** (0.034) [0.025]**	-0.251*** (0.080) [0.015]**	0.118* (0.070) [0.054]*
Control mean	0.221	0.379	0.075	0.312	0.087	0.221	0.458	0.087
Observations	554	554	95	95	95	554	554	93

Notes: **Table A39 shows treatment effects on post-treatment behavior including control variables.** It displays the effects of the vacancy subsidy intervention on post-treatment vacancy posting and hiring in the two months following the four-month treatment period. Columns 1 to 5 show treatment effects on vacancy creation. Column 1 shows the effects on creating any vacancy post-treatment. Column 2 shows the effect on the number of created vacancies. Column 3 shows the effect on the fraction of formally posted vacancies. Column 4 shows the effect on the fraction of vacancies searched through networks. Column 5 shows the effect on the fraction of white-collar vacancies. Columns 6 to 8 show effects on hiring outcomes. Column 6 shows the effect on having any hire post-treatment. Column 7 shows the effect on the number of post-treatment hires. Column 8 shows the effect on the fraction of white-collar hires. Outcomes in columns 3, 4, 5, and 8 are only defined for firms with at least one created vacancy or hire. White-collar vacancies are defined as “professional, managerial, or administrative” workers. Typical white-collar job titles include manager, accountant, and supervisor. Typical non-white-collar jobs are cooks, waiters, or carpenters. All specifications include control variables selected from a pre-specified set of variables using LASSO algorithms for each outcome separately. Heteroskedasticity robust standard errors are displayed in parentheses. Minimum q-values from two-stage false discovery rate correction are displayed in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Additional figures

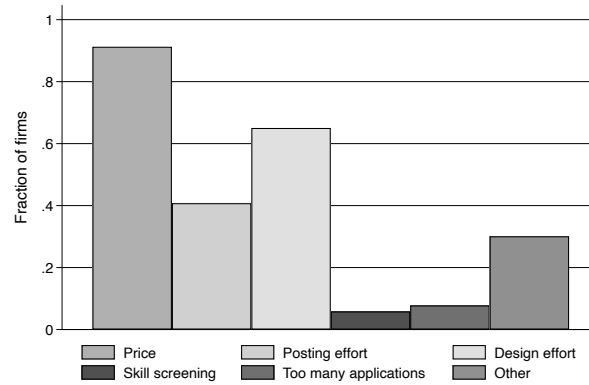
Figure A1: Geographical distribution of firms



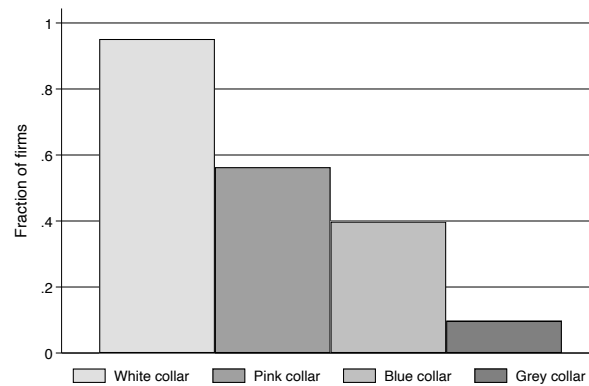
Notes: Figure A1 displays the spatial distribution of sampled firms across Addis Ababa, Ethiopia.

Figure A2: Firms' perceptions of formal search

(a) Stated obstacles to using formal hiring channels

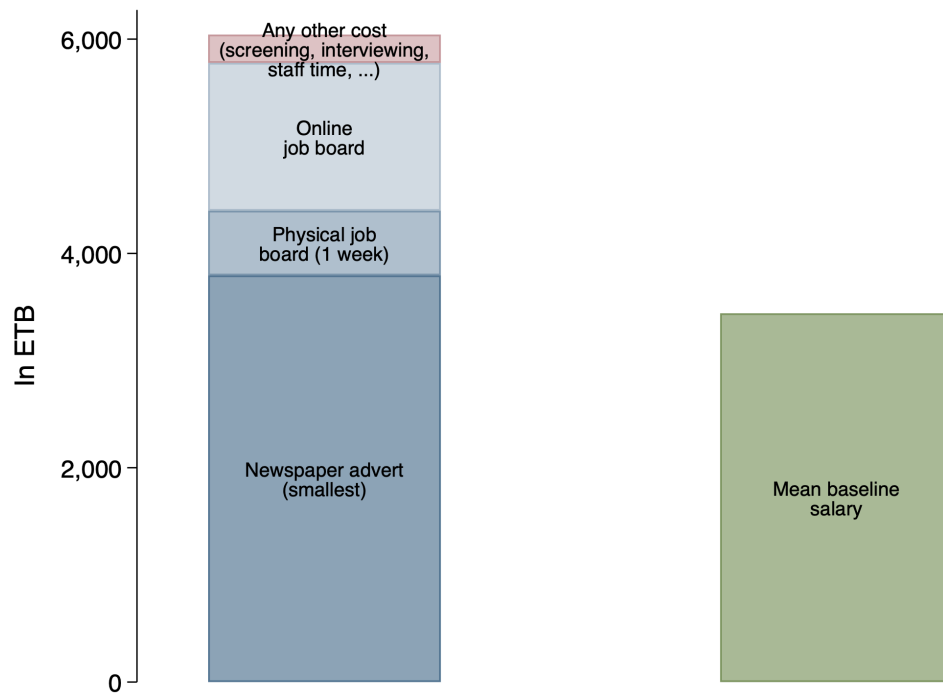


(b) Stated usefulness of formal hiring channels by position type



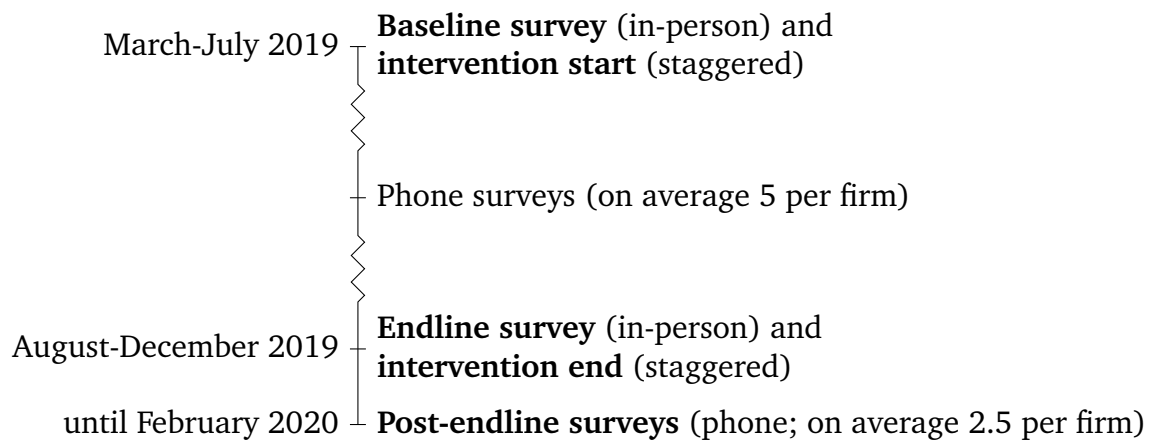
Notes: Figure A2 shows that managers perceive formal search as expensive and most suitable for white-collar workers. Figure A2a displays the fraction of firms that mention a given factor in an open question asking for the “main difficulties in using formal hiring channels”. Each manager could mention multiple factors. Figure A2b displays managers’ responses to the following question: “For the hiring of what type of worker are formal search channels most useful?” Managers could select one or more of four job types: white-/ blue-/ pink-/ gray-collar workers. Typical white-collar/professional job titles include manager, accountant, and supervisor. Typical blue-collar or production jobs are machine operators, builders, factory floor workers, and cooks. Typical pink-collar/service jobs are waiters, tellers, shop attendants, and hotel concierges. Typical gray-collar/ support workers are guards, drivers, cleaners, tea makers, and couriers/messengers. Data was collected in 2023 among 103 randomly sampled control group firms of the original experiment in 2019.

Figure A3: Cost of the vacancy-posting subsidy



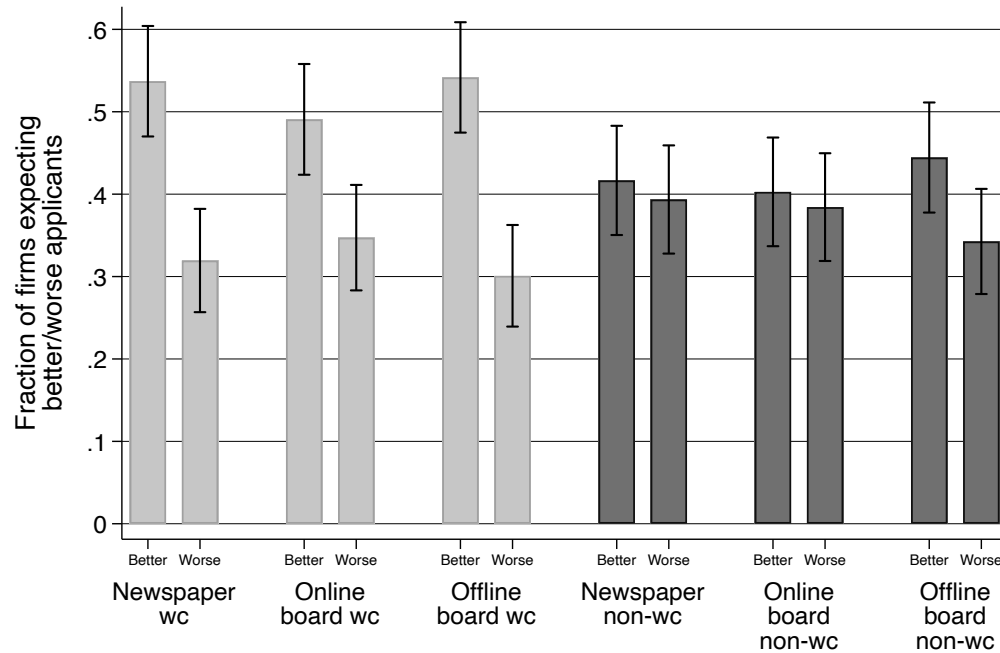
Notes: **Figure A3 compares the cost of a formal vacancy posting to firms' baseline hiring costs and average salary levels.** The left bar reports the cost of posting a vacancy through the bundle of formal channels used in the intervention: a newspaper advertisement, a physical job board posting for one week, and an online job board listing. The top segment labeled "any other cost" reports average baseline non-posting hiring costs, including screening, interviewing, and staff time. The right bar reports the mean baseline salary. Costs are measured in Ethiopian Birr (ETB). The figure illustrates that formal vacancy posting was costly relative to firms' baseline hiring expenditures and average wage levels.

Figure A4: Timeline



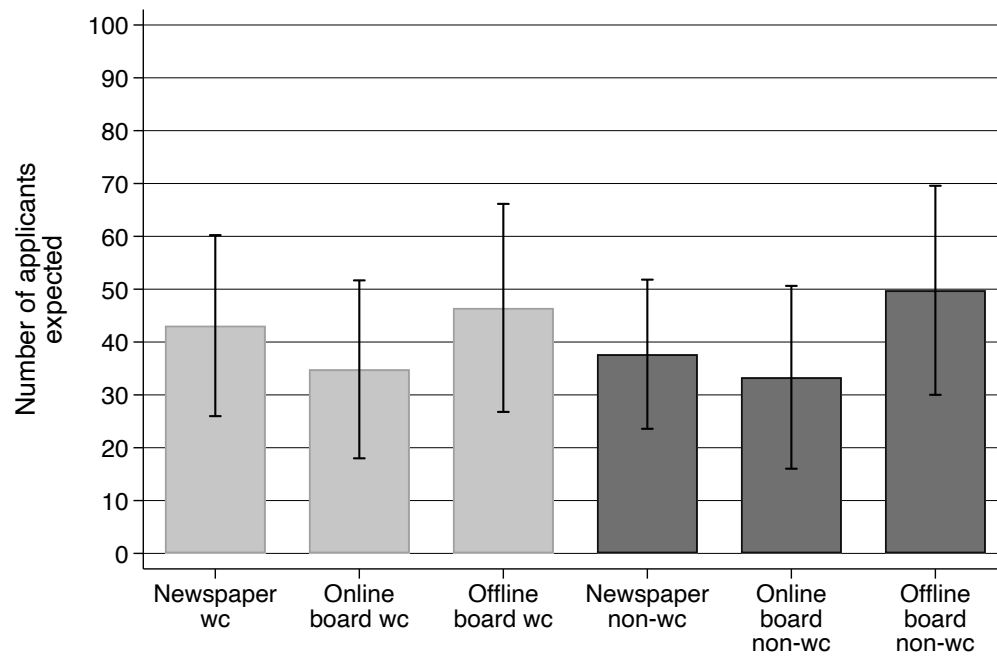
Notes: Figure A4 displays the timeline of the experiment.

Figure A5: Firm expectations about relative quality of formal applicants



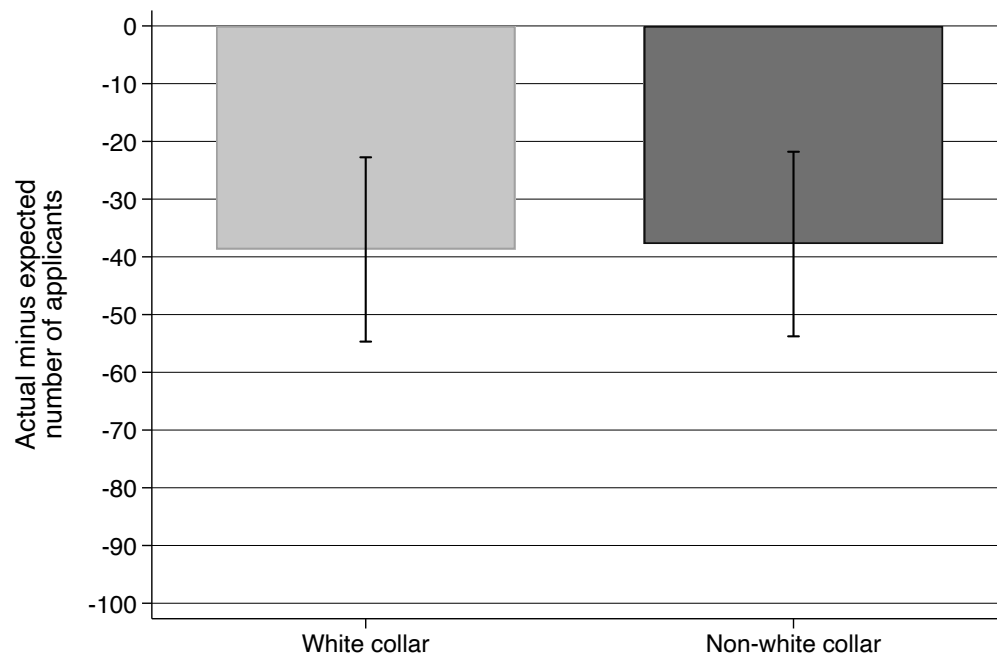
Notes: **Figure A5 shows that managers perceive formal channels to be more useful for white-collar positions.** It shows the share of firms who expect applicants to vacancies posted in formal channels to be of better or worse quality compared to applicants obtained through social networks. The figure is shown for control group firms only, based on endline data. The three formal channels are newspapers, online job boards, and offline job boards. In every pair of bars, i) the left bar shows the share of firms that expect applicants obtained through the respective channel to be of better quality than those obtained through social networks (based on the top three responses on a seven-point Likert scale question); ii) the right bar shows the share of firms that expect these applicants to be of worse quality. The three left pairs of bars show expectations over applicants to white-collar positions, the three right pairs of bars show expectations over applicants to non-white-collar positions.

Figure A6: Firm expectations about applicant numbers



Notes: **Figure A6 shows that the number of applicants firms expect to receive when posting a vacancy in a given formal channel is relatively high.** The numbers are substantially higher than the ones observed in Table A11. The figure is based on control group endline data. The three formal channels are newspapers, online job boards, and offline job boards. The three left bars show expectations over applicant numbers to white-collar positions, the three right bars show expectations over applicant numbers to non-white-collar positions.

Figure A7: Difference between actual and expected number of applicants



Notes: **Figure A7 shows that firms are overoptimistic about the number of applicants they can attract through formal channels.** It shows the firm-level difference between actual and expected number of applicants, by collar type. The figure is shown for control group firms only, based on endline data. The left bar shows the difference for applicant numbers to white-collar positions, the right bar shows the difference for applicant numbers to non-white-collar positions.

Figure A8: Sample candidate skill report

Applicant summaries for [REDACTED]

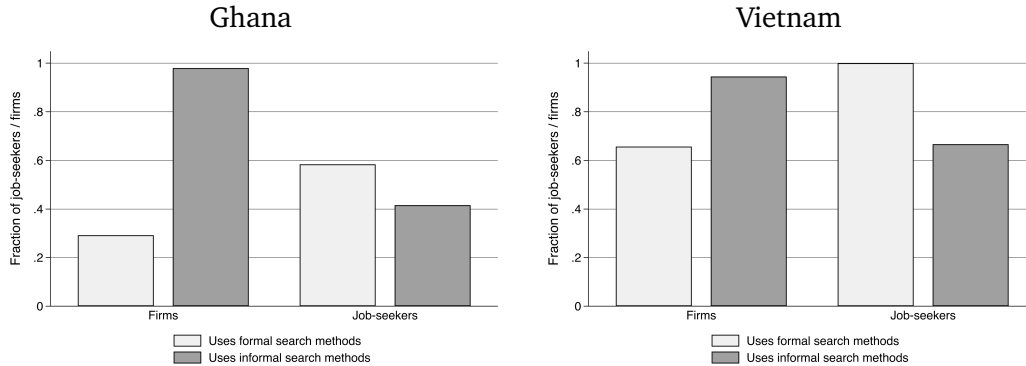
The Policy Studies Institute (PSI) and the University of Oxford have offered an applicant screening service to your firm.

For your vacancy for an Accountant (vacancy number 10131002), the following individuals applied for the job. We have ranked them below based on their overall score, starting with the applicant who fits best to the vacancy. Applicants to this position are divided into three groups. Applicants in the top third are marked green. Applicants in the middle third are marked yellow. Finally, applicants in the bottom third are marked red.

Rank	Name	Overall Score	General Intelligence	Reliability	Trust in others	Contact
		Lo ----- medium ----- hi	Lo ----- medium ----- hi	Lo ----- medium ----- hi	Lo ----- medium ----- hi	
1	[REDACTED]	[Green bar]	[Yellow bar]	[Green bar]	[Green bar]	[REDACTED]
2	[REDACTED]	[Green bar]	[Green bar]	[Green bar]	[Yellow bar]	[REDACTED]
3	[REDACTED]	[Green bar]	[Green bar]	[Green bar]	[Red bar]	[REDACTED]
4	[REDACTED]	[Green bar]	[Green bar]	[Yellow bar]	[Yellow bar]	[REDACTED]
5	[REDACTED]	[Green bar]	[Green bar]	[Yellow bar]	[Yellow bar]	[REDACTED]
6	[REDACTED]	[Green bar]	[Yellow bar]	[Green bar]	[Yellow bar]	[REDACTED]
7	[REDACTED]	[Green bar]	[Yellow bar]	[Yellow bar]	[Yellow bar]	[REDACTED]
8	[REDACTED]	[Green bar]	[Green bar]	[Yellow bar]	[Yellow bar]	[REDACTED]
9	[REDACTED]	[Green bar]	[Green bar]	[Red bar]	[Yellow bar]	[REDACTED]
10	[REDACTED]	[Yellow bar]	[Green bar]	[Yellow bar]	[Yellow bar]	[REDACTED]
11	[REDACTED]	[Yellow bar]	[Green bar]	[Yellow bar]	[Red bar]	[REDACTED]
12	[REDACTED]	[Yellow bar]	[Red bar]	[Green bar]	[Yellow bar]	[REDACTED]

Notes: Figure A8 displays the first page of a sample candidate skill report provided to firms in the subsidy plus screening group. Reports include information about the three out of ten skills firms select as most important. Scores are based on an in-person assessment of applicants to vacancies submitted by firms in the subsidy plus screening treatment group. The ten skills and character traits firms can choose from are General Intelligence (Raven Matrices, [Raven \(2000\)](#)), Emotional Intelligence (Reading the Mind in the Eyes test, [Baron-Cohen et al. \(2001\)](#)), Mathematical Ability (Middle School curriculum-based test), English language ability (Middle School curriculum-based test), Grit/Persistence (Grit scale, [Duckworth et al. \(2007\)](#)), Reliability (sum of standardized variables: a dummy of being on time, a dummy of sending a confirmation message before the screening, and Big Five conscientiousness, [John and Srivastava \(1999\)](#)), Emotional Stability (inversely coded Big Five Neuroticism, [John and Srivastava \(1999\)](#)), Focus (sum of standardized Stroop test variables: number of correct responses, and time taken (reverse-coded), [Stroop \(1935\)](#)), Trust in others (first stage of incentivized trust game, [Berg et al. \(1995\)](#)), and Reciprocity (sum of standardized variables: second stage of incentivized trust game ([Berg et al., 1995](#)), and negative reciprocity in an ultimatum game ([Fehr and Gächter, 2000](#))). For each skill, applicants are ranked in terciles within screened applicants to a given vacancy. Ties are broken in favor of the higher tercile. The aggregate candidate ranking is generated based on the sum of standardized scores of the three selected skills. Overall terciles are calculated within vacancy based on the same sum. Firms receive one report per vacancy, either printed out or via email depending on firm preferences. Firm and candidate details have been redacted.

Figure A9: Formal and informal search on both sides of the labor market in Ghana and Vietnam

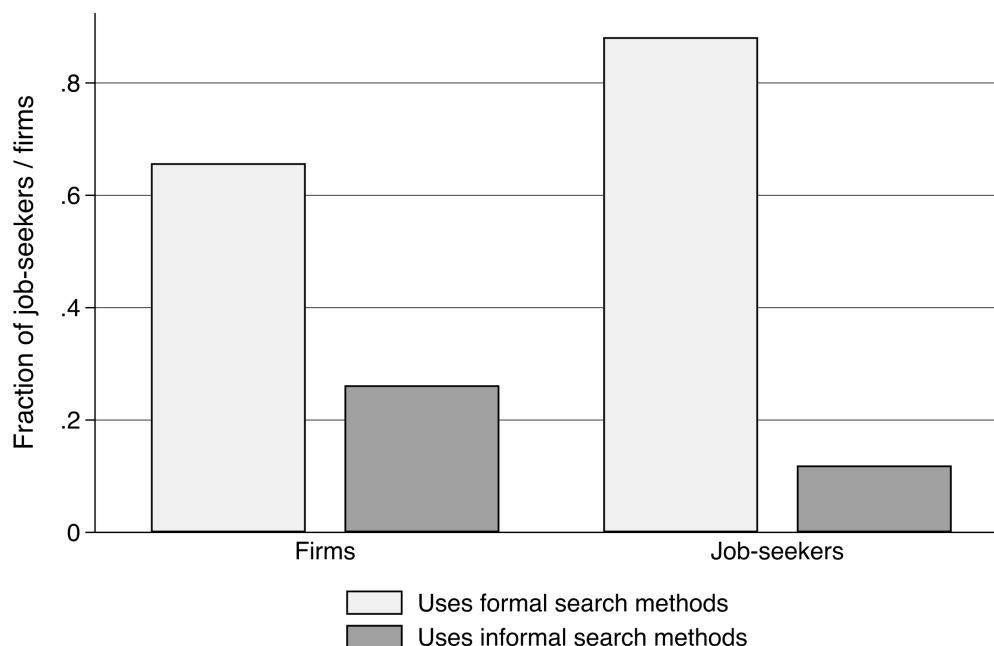


Notes: Figure A9 compares formal and informal search by firms and job-seekers in Ghana and Vietnam. Each panel shows the fraction of firms and job-seekers relying on formal or informal search methods. The responses are not mutually exclusive, so both firms and job-seekers may use more than one channel.

In the Ghana panel, the firm data are from the Ghana 2007 World Bank Enterprise Survey and the job-seeker data are from the urban sample reported in Table 5.17 of the Ghana Labor Force Survey 2015. On the firm side, formal search is proxied by external recruitment and informal search by non-external recruitment. On the job-seeker side, informal search is asking friends and relatives, while formal search includes applying to prospective employers, newspaper advertisements, the internet, and employment services.

In the Vietnam panel, the firm data are from the Vietnam STEP Employer Survey, restricted to firms with 50 or fewer employees, and the job-seeker data are from the Vietnam STEP Household Survey. On the firm side, formal search includes public or private employment services, job fairs, educational institutions, media advertisements, and internet recruitment; informal search is recruitment through informal channels. On the job-seeker side, the sample is restricted to urban, younger, and more educated active wage-job-seekers. Formal search includes all non-network methods, while informal search is search through social networks.

Figure A10: Formal and informal search on both sides of the labor market in Malaysia



Notes: **Figure A10 compares formal and informal search by firms and job-seekers in Malaysia.** It shows the fraction of firms and job-seekers relying on formal or informal search methods. Firms and job-seekers may use more than one channel. The firm data are from the Malaysia 2007 World Bank Enterprise Survey. On the firm side, formal search is defined as firms relying predominantly on formal recruitment channels, including factory-door postings, newspaper advertisements, headhunters, and the electronic labor exchange, while informal search is defined as relying predominantly on referrals through friends or other workers. The job-seeker data are from Table 6.8A of the Department of Statistics Malaysia Labor Force Survey 2023. On the job-seeker side, informal search is search through relatives and friends, while formal search includes employment or labor agencies, responding to advertisements, and sending applications. The figure shows a contrast case to Ethiopia: both firms and job-seekers in Malaysia rely predominantly on formal search methods.

B Sample representativeness

This section compares our sample to representative samples from Addis Ababa. Panel A of Table B1 shows firm-level statistics comparing data collected in 2020 by the Addis Ababa Labor, Enterprise, and Industry Development Bureau (AALEID) to our sample. The AALEID data is a complete data set of

micro-, small-, and medium-sized active firms in Addis Ababa. Larger firms are under the auspices of federal agencies.³³ Panel B shows aggregated employment share and salary statistics based on the Urban Employment Unemployment Survey in 2018 and the Labor Force and Migration Survey of 2020, both of which are conducted by the Central Statistical Agency. This data source is representative at the worker level, and covers employment at firms of all sizes. We focus our descriptive analysis on the subsample of wage employed individuals.

Our sample falls, by design, into the upper tail of the size distribution of the AALEID sample. 63% of firms are micro-enterprises with five or fewer employees, whereas 37% have between 6 and 50 employees.³⁴ Our sample consists of SMEs with between 5 and 50 employees, with only two firms reporting more than 50 employees and one firm reporting 4 employees. The small discrepancy to our screening criteria stems from hiring between pre-screening and baseline surveys.

The manufacturing sector makes up 28% of all AALEID firms and 39% of small- and medium-sized firms in Addis Ababa, but it only employs 13% of workers. Our sample consists of 51% manufacturing firms, which comprise 47% of employment in our sample. The service sector makes up 76% of AALEID firms and only 36% of small- and medium-sized firms in Addis Ababa. In our sample, 38% of firms work in the service sector. The employment share of the service sector in Addis Ababa is 76%, substantially more than the 31% in our sample.

Firms in our sample employ 13% white-collar workers at baseline, which is somewhat less than the share of 21% white-collar positions among wage employed individuals. This suggests that our sample employs relatively fewer high-skilled workers.

³³There is, to our knowledge, no representative firm data from Addis Ababa that covers all sectors and firm sizes. The 2015 World Bank Enterprise survey of small, medium, and large firms documents a ratio of one large firm with more than 50 employees per four firms with between 5 and 50 employees.

³⁴Extrapolating from the World Bank Enterprise Survey yields 58% micro enterprises, 33% SMEs, and 9% large enterprises.

Finally, our sample is somewhat positively selected in terms of workers' salaries. The average worker at firms in our sample is paid 3779 ETB per month whereas workers in Addis Ababa earn 3086 ETB per month. The selection differs by sector: workers at service sector firms earn less than in the representative sample, whereas workers in the manufacturing sector earn slightly more than in the representative sample.

When splitting the sample by occupation, we find positive selection for both white-collar and non-white-collar workers. This implies that the over-optimistic expectations of applicants documented in section 6.1 are unlikely to be the result of negative selection of firms into our experimental sample.

Table B1: Representativeness of sample

Panel A: Firm Characteristics			
Share of firms	AALEID data	Among SMEs	Our Sample
Micro Enterprises	63%		0%
Small and Medium Enterprises	36%		100%
Manufacturing Sector	28%	39%	51%
Service Sector	40%	34%	38%
Other	33%	32%	11%

Panel B: Labor market data		
Employment Shares	Repr. Data	Our Sample
Share Manufacturing	13%	47%
Share Service Sector	76%	31%
Share White-Collar	21%	13%
Share Non-White-Collar	79%	87%

Monthly salaries (ETB)		
All	3086	3779
Manufacturing	3385	4183
Service	3401	2219
White-Collar	5432	6021
Non-White-Collar	2479	3436

Notes: **Table B1 compares our sample to other data sources.** Panel A compares our sample to data on active firms with up to 50 employees from the Addis Ababa Labor, Enterprise, and Industry Development Bureau (AALEID). Micro enterprises are defined as 1 to 4 workers. SMEs are defined as 5 to 50 employees. Panel B uses data from the Urban Employment Unemployment Survey in 2018, for wages and occupations, and data from the Labor Force and Migration Survey of 2020 for sectoral composition.

C Vacancy characteristics

This Appendix Section describes the differences between white-collar and non-white-collar vacancies based on our representative sample of publicly posted vacancies in Addis Ababa in 2019. Table C1 shows that almost all white-collar vacancies require applicants to have an education beyond a high school degree. Almost 60% of white-collar vacancies require a university degree and the remaining 40% require vocational training. The educational requirements of non-white-collar vacancies are much lower. A quarter of non-white-collar vacancies require at most high school education, while only 35% require a university degree. Similarly, white-collar vacancies require, on average, almost a year (or 38%) more experience than non-white-collar vacancies. Lastly, based on the selective subsample of vacancies for which firms decided to post salaries, white-collar vacancies, on average, pay two and a half times higher monthly salaries than non-white-collar vacancies (a 148% increase). Note that this comparison only considers formally posted vacancies. We do not directly observe differences between informally posted white-collar and non-white-collar vacancies but expect qualitatively similar differences among informally posted vacancies.

Taken together, we find that white-collar vacancies have relatively high education and experience requirements. They also offer substantially higher wages than non-white-collar vacancies. This pattern suggests that white-collar vacancies are more attractive and more demanding than non-white-collar vacancies. Considering that white-collar vacancies are more likely to be posted formally, this also implies that formally posted vacancies are positively selected.

Table C1: Differences between white-collar and non-white-collar vacancies

	White-collar (WC)	Non-white- collar (NWC)	Δ	p(WC=NWC)	Obs.
Number of vacancies	593	406	187	.	999
Share of vacancies	0.59	0.41	0.19	.	999
Education requirements					
High school or less	0.01	0.25	-0.23	0.00	999
Vocational/other post-secondary	0.40	0.41	-0.01	0.87	999
University	0.59	0.35	0.24	0.00	999
Other characteristics					
Years of experience	3.28	2.38	0.90	0.00	877
Monthly salary	10086	4052	6034	0.00	134

Notes: Table C1 compares various vacancy characteristics between white-collar and non-white-collar vacancies. The monthly salary in row five is in Ethiopian Birr (ETB). Based on a random subsample of our database of publicly posted vacancies in Addis Ababa in 2019, where we coded the collar details of a random sample of $N = 999$ vacancies. The differences in number of observations from the total sample of $N = 999$ are due to missing experience requirements or salary information.

D Wage expectation and formal search

In this Section we explore how the differences between wage expectations and realized earnings differ between job-seekers who search formally and those who do not. To do so, we use data on 1245 control group job-seekers in Addis Ababa in 2022 from [Tekleselassie et al. \(2025\)](#). The authors conduct a field experiment recruiting job-seekers through job adverts distributed through a range of channels (offline and online job boards and social media) and then screen applicants in a similar set-up to this experiment. At baseline, we find that 59.4% of job-seekers engage in formal job search. We also observe that applicants who search in formal channels have 11.5% higher wage expectations and 11.9% higher reservation wages. Both differences are highly statistically significant with $p = 0.001$ and $p = 0.002$, respectively. This is the first suggestive evidence that applicants obtained through formal search channels have higher wage expectations.

Next, we assess expectation gaps and compare these beliefs with realized monthly earnings about six weeks later. We first divide job-seekers into 32 groups based on baseline formal search, targeted vacancy type (white-collar, blue-collar, pink-collar, gray-collar), gender, and a dummy indicating university education. We then calculate average monthly earnings among working individuals within each of these groups. We calculate the expectation gap as the difference between individually expected wages and realized earnings in the group the job-seeker belongs to and normalize it by group-specific realized earnings. We find that wage expectations are significantly higher than realized earnings: wage expectations are, on average, 43.6% higher than realized wages while reservation wages are, on average, 32.8% higher than realized wages (both $p < 0.001$).

Importantly, the expectations gap is substantially larger among those who engage in formal search at baseline. Wage expectations among those who did not search formally are only 25% higher than realized earnings, while those who search formally have expectations that are 56.7% higher than realized earnings. For reservation wages, we find that the gap is 15.5%

Table D1: Differences in wage expectations levels and expectation gaps by whether job-seekers search formally

	Level (ETB)		Expectation gap (%)	
	(1)	(2)	(3)	(4)
	Reservation wage	Wage expectation	Reservation wage	Wage expectation
Any formal search	770.1** (255.9)	812.5** (281.9)	0.294*** (0.0529)	0.317*** (0.0558)
Constant	6472.4*** (192.8)	7062.9*** (216.7)	0.155*** (0.0367)	0.250*** (0.0391)
Observations	1274	1274	1254	1254

Notes: **Table D1 explores differences between wage expectations and realized earnings by whether job-seekers search formally.** The sample consists of control group job-seekers in Addis Ababa in 2022 from [Tekleselassie et al. \(2025\)](#). At baseline 59.4% of job-seekers used formal search channels. Formal search channels are defined as physical or online job boards and newspapers. Column 1 shows differences in baseline reservation wages in Ethiopian Birr (ETB). Column 2 shows differences in baseline wage expectations in Ethiopian Birr (ETB). Columns 3 and 4 show differences in expectation gaps in percent. The expectation gap is calculated as the difference between individual wage expectations and realized earnings about six weeks later. To estimate realized earnings we first divide job-seekers into 32 groups based on baseline formal search, targeted vacancy type (white-collar, blue-collar, pink-collar, gray-collar), gender, and a dummy indicating university education. We then calculate average monthly earnings among working individuals within each of these groups. We calculate the expectation gap as the difference between individually expected wages and realized earnings in the group the job-seeker belongs to and normalize it by group-specific realized earnings. 100 ETB were equivalent to 5.79 USD PPP in 2022. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

among those not searching formally and 44.8% for those searching formally. The differences in expectation gaps are significant with $p < 0.001$ for both measures.

This pattern is entirely consistent with the expectation gap mechanism we propose in Section 6. Job-seekers using formal search channels overestimate feasible earnings more than those who do not, leading to lower vacancy filling rates for formally posted vacancies.

E Theoretical framework

This Section lays out a theoretical framework that rationalizes the descriptive patterns shown in Section 2 and makes predictions about treatment effects. It describes firms' decisions of whether to create a vacancy, how to advertise it, and what type of position to fill.

E.1 Setup

Consider a firm deciding whether to attempt to hire a new employee. Firm i chooses the optimal search strategy s_i^* and the vacancy type t_i^* to maximize the expected profits:

$$(s_i^*, t_i^*) = \operatorname{argmax}_{s_i, t_i} \pi(s_i, t_i) - c(s_i) \quad (3)$$

where $\pi(s_i, t_i)$ is firm i 's expected profit of searching for an employee of type t (white-collar (*wc*) or non-white-collar positions (*nwc*)) through channel s_i (formal or informal)³⁵ excluding the search cost. $c(s_i)$ are the costs of searching for employees in search channel s_i .

Firms' expected profit from search The expected profit from employee search $\pi(s_i, t_i)$ varies across search channels and vacancy types. We assume that the returns to informal search are higher for non-white-collar vacancies compared to white-collar vacancies ($\pi(\text{informal}, wc) < \pi(\text{informal}, nwc)$). This assumption is in line with non-white-collar vacancies being posted informally at much higher rates than white-collar vacancies.

³⁵For simplicity, we assume that these two are mutually exclusive.

Cost of search Firms' costs of searching for applicants in channel s_i have two components: $c(s_i) = c^{adv}(s_i) + c^{scr}(s_i)$, where $c^{adv}(s_i)$ are costs of advertising the position on channel s_i , and $c^{scr}(s_i)$ are the total costs of screening applicants obtained through channel s_i , to learn the applicants' true quality.³⁶ We assume that formal search is more costly than informal search: $c(formal) > c(informal)$. This reflects both higher advertising cost $c^{adv}(formal)$ and, potentially, higher screening cost $c^{scr}(formal)$, e.g. because informal network search can provide more information about applicants.

Choosing the search channel We first consider firms' choice of search channel conditional on job type t . Firm i chooses formal search for job type t if the expected profit from doing so is larger than the expected profit from searching informally:

$$\pi(t, formal) - \pi(t, informal) > c(formal) - c(informal) = \Delta c \quad (4)$$

That is, firms will use formal job search for type t if the difference in expected benefit between formal and informal search is larger than the cost differential between the two search methods. This condition yields two thresholds for the profitability of formal search below which firms would prefer informal search channels for type t : $\pi^*(wc, formal) = \Delta c + \pi(wc, informal)$ and $\pi^*(nwc, formal) = \Delta c + \pi(nwc, informal)$.

Choosing the vacancy type Next, we consider firms' optimal choice of vacancy type. Firms choose the vacancy type to maximize expected profits, taking into account the optimal search strategy for each type of job $s^*(t)$. They choose a white-collar vacancy if:

$$\pi(s^*(wc), wc) - c(s^*(wc), wc) > \pi(s^*(nwc), nw) - c(s^*(nwc), nw) \quad (5)$$

We can combine this condition with the choice of search channel to completely characterize firms' search behavior.

³⁶For simplicity, we assume that screening is always optimal for firms.

Combining type and search choice Figure E1 displays firms' optimal choices in the $\pi(wc, formal) - \pi(nwc, formal)$ space.

First, if it is optimal to use informal search channels for both job types ($\pi(t_i, formal) < \pi^*(t_i, formal), \forall t_i$), firms' type decision will only depend on their profitability of informal search $\pi_i(informal, t)$. Hence, firms will choose a non-white-collar vacancy.³⁷ This corresponds to area A in Figure E1.

Second, if formal search is optimal for all types, that is $\pi(t_i, formal) > \pi^*(t_i, formal), \forall t_i$, firms will post a white-collar vacancy if $\pi(wc, formal) > \pi(nwc, formal)$ (area F). Otherwise they will choose to post a non-white-collar vacancy formally (area E).

Third, if formal search is optimal for white-collar jobs but not for non-white-collar jobs ($\pi(wc, formal) > \pi^*(wc, formal)$ and $\pi(nwc, formal) < \pi^*(nwc, formal)$; areas B and C), we need to compare the profits across the optimal search channels. Firms will post white-collar vacancies if $\pi(wc, formal) > \pi(nwc, informal) + \delta c = \pi^*(nwc)$ (area C). If $\pi^*(nwc) > \pi(wc, formal)$ firms will post informal non-white-collar vacancies (area B). Hence, both formal white-collar and informal non-white-collar posting will occur in this regime.

Finally, if formal search is optimal for non-white-collar jobs but not for white-collar jobs, firms will only create non-white-collar vacancies (that is $\pi(wc, formal) > \pi^*(wc, formal)$ and $\pi(nwc, formal) < \pi^*(nwc, formal)$; area D). Firms would post white-collar vacancies if it were profitable, that is $\pi(nwc, formal) < \pi(wc, informal) + \delta c = \pi^*(wc)$. However, to be in this search regime we require $\pi(nwc, formal) > \pi^*(nwc)$. As we assume $\pi(wc, informal) < \pi(nwc, informal)$ we know that $\pi^*(wc) < \pi^*(nwc)$, which implies that firms will not post white-collar vacancies in this regime.

The dashed line in Figure E1 summarizes this pattern. If firms fall above the dashed line, they will choose to create a white-collar vacancy (areas C and F). Conversely, if firms fall below the dashed line, they will choose

³⁷Remember that we assumed $\pi(wc, informal) < \pi(nwc, informal)$, i.e. that non-formal search has higher returns for non-white-collar vacancies.

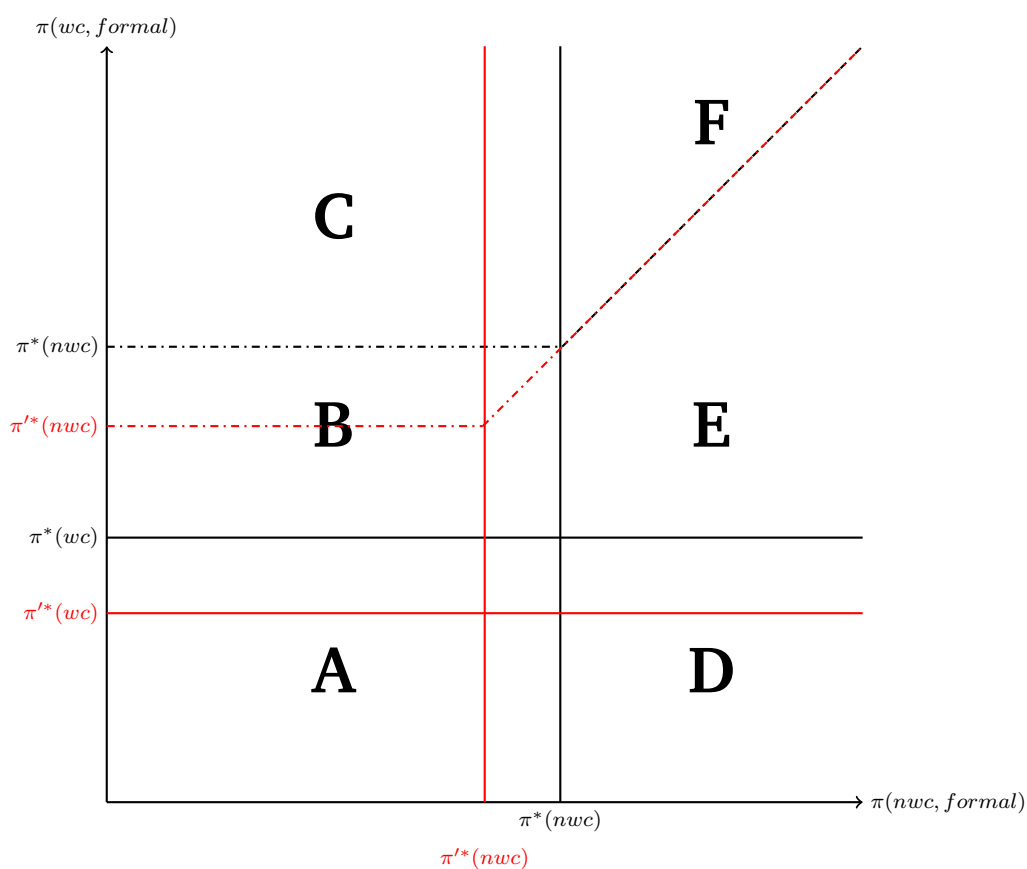


Figure E1: Vacancy search and type decision

Notes: Figure E1 displays the choice of search channel and vacancy type in the $\pi(nwc, formal)$ — $\pi(wc, formal)$ space. Red lines indicate what happens when formal search costs are reduced. Firms that fall in the area above the dashed lines post white-collar vacancies (areas C and F). Firms that fall below the dashed line post non-white-collar vacancies (areas A, B, D, and E).

to create a non-white-collar vacancy (areas A, B, D, and E). All white-collar vacancies will be posted formally (areas C and F), while only a subset of non-white-collar vacancies will be posted formally (areas D and E). Firms that fall in areas A and B will search informally for non-white-collar employees.³⁸

Vacancy creation decision Finally, firms decide whether or not to create a vacancy in the first place. They do so if the expected return is positive:

$$\max_{t,s} \pi(t_i, s_i) - c(s_i) > 0 \quad (6)$$

This implies that the vacancy creation decision depends on the anticipated vacancy type and search channel.

E.2 Subsidizing formal search

Our field experiment effectively subsidizes the cost of formal search. In the model, this means that $c'(formal) < c(formal)$. This implies that the cost differential between formal and informal search channels is also reduced ($\Delta c' < \Delta c$). The red lines in Figure E1 display the effect of the subsidy graphically. We derive four predictions from the model that we will test empirically using our experimental data.

Result 1: Subsidies increase take-up of formal search. This result follows from the intuition that reducing the price of formal search will allow firms with lower benefits from formal search to use formal search. Graphically, we observe that the areas C+F and E+D, which contain all formal postings, grow unambiguously larger. This is the case because the subsidy shifts both formal posting thresholds $\pi^*(wc) = \Delta c + \pi(wc, informal)$ and $\pi^*(nwc) = \Delta c + \pi(nwc, informal)$ closer to the origin as Δc decreases. Moreover, this subsidy will not dissuade any firms from creating vacancies as none of the costs increase. Hence, subsidizing formal search will make firms with lower benefits from formal search choose formal search. However, the model also implies that there will still be firms who prefer to use informal search

³⁸If the returns to informal search were higher for white collar vacancies ($\pi(wc, informal) > \pi(nwc, informal)$), all non-white-collar vacancies would be posted formally and some white-collar vacancies would be posted informally (see Figure E2).

channels for their vacancies, implying that take-up will not be perfect.

Result 2: Subsidies increase the number of created vacancies. Introducing vacancy subsidies can never decrease the expected profits, as it either reduces cost or leaves cost unchanged. Consider firms who are close to indifferent between creating a formally advertised vacancy and not posting at all but optimally decide to not create a vacancy without the subsidy. Such firms will be induced to create a vacancy as the costs of formal search decrease.³⁹ The magnitude of the effect depends on the mass of firms that fall into this category. If there are many firms who are close to being indifferent between posting a vacancy formally and not posting a vacancy, we would expect relatively large effects of the subsidy. However, if most firms without initial vacancies are far from the threshold values for formal search, we would expect relatively small treatment effects.

Result 3: Subsidies lead to an increase in white-collar vacancies. The vacancy subsidy increases white-collar vacancy posting by making formal posting more attractive. This leads to some informal non-white-collar vacancies being substituted by formally posted white-collar vacancies. This happens as the range of $\pi(t_i, s_i)$ for which firms post informal non-white-collar vacancies rather than formal white-collar vacancies (area B in Figure E1) shrinks because formal posting is more profitable for both white-collar and non-white-collar vacancies. Put differently, the area in which formal white-collar posting is optimal (area C+F in Figure E1) increases as the cost of formal search decreases. As there are no informally posted white-collar vacancies in the model, the additional formal white-collar vacancies substitute non-white-collar vacancies. This effect is driven by firms with relatively

³⁹Formally, consider a firm for which formal search for white-collar is most profitable ($\pi(wc, formal) - c(formal) \geq \pi(t, s) - c(s) \forall t_i, s_i$). This firm will post a vacancy if $\pi(wc, formal) - c(formal) > 0$. Define $\bar{\pi}_i(wc, formal) = c(formal)$ as the threshold value above which this firm posts a vacancy. The vacancy subsidy reduces this threshold to $\bar{\pi}'_i(wc, formal) = c'(formal)$. Now, other firms with $\bar{\pi}_i(wc, formal) > \pi(wc, formal) > \bar{\pi}'_i(wc, formal)$ and for which $\pi(wc, formal) - c(formal) \geq \pi(t_i, s_i) - c(s_i) \forall t_i, s_i$ holds will be induced to post formal white-collar vacancies.

high returns to formal search for white-collar compared to non-white-collar vacancies (i.e. firms that fall above the 45 degree line in Figure E1).⁴⁰

Result 4: Reducing screening costs can further amplify the effect of subsidies. This result directly follows from the above results. If we further reduce the screening cost that firms face when using formal search channels ($c^{scr}(formal)$), we further reduce the cost of formal search to $c''(formal) < c'(formal) < c(formal)$. If the cost of screening is a constraining factor for firms, then we would expect the additional reduction to lead to additional uptake of formal search. It could also imply a further shift to white-collar vacancies following the same logic as in Result 3.

Beliefs about returns to formal employee search So far, we have assumed that the firms' managers have full information about the expected returns to search through different channels ($\pi(t_i, s_i)$). However, in practice that may not be the case. The overwhelming prevalence of informal search and low demand for white-collar employees means that few managers might have direct experience in using these channels, which could help them form accurate beliefs. Indeed, Chandrasekhar et al. (2020) show that—in the presence of informal search—even rational firms might fail to learn about the true returns to formal search.

If firm managers are too pessimistic about the returns to formal search, subsidizing formal search could lead to learning and permanent behavioral change. Conversely, if managers are overoptimistic about the returns to formal search, they might attempt to use subsidized formal search initially. The overoptimism might lead to unsuccessful searches, for example, because managers have unrealistic expectations about the quality and quantity of applicants. This could lead to reduced hiring rates among treated firms. If this were the case, we would expect firm managers to update their beliefs and revert back to informal search methods in the long run.

⁴⁰Subsidies could also increase white-collar vacancies by causing non-posting firms to start posting white-collar vacancies as argued for Result 2.

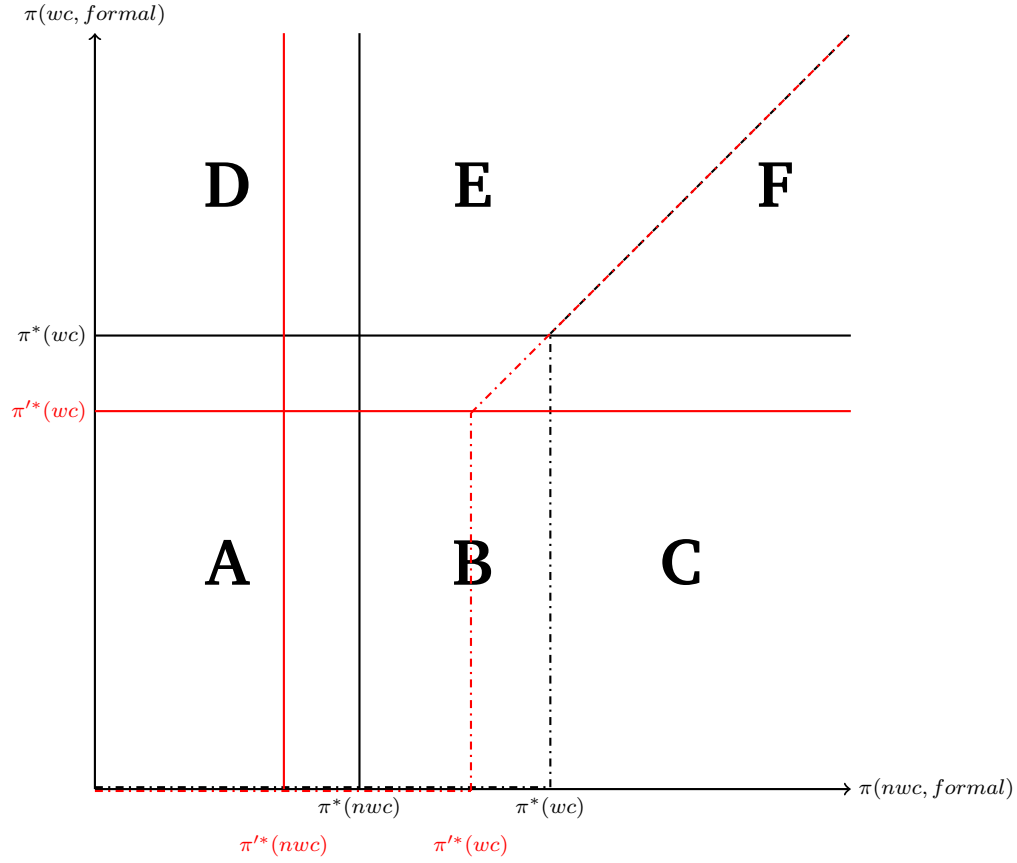
F Deviation from the pre-analysis plan

In our analysis, we make the following deviations from our pre-analysis plan:

- We expand the number of outcomes to study both the extensive and intensive margins of vacancy creation, as well as the ratio of filled vacancies to overall vacancies. To accommodate the larger number of outcomes, we spread these across multiple tables and correct for multiple hypothesis testing across the different margins.
- We collapse all different non-white-collar employment categories (blue collar, pink collar, gray collar) into a single non-white-collar category to improve power. We also collapse managers' expectations in the same way.
- We do not normalize outcomes over time, in order to be able to use extensive margin outcomes.
- We do not winsorize the number of vacancies and number of hires because there are no outliers.
- For our main specification, we estimate pooled treatment effects instead of separate effects for a screening add-on intervention, because we do not find consistent differences across treatment arms.
- We only show firm-level regression specifications. We do not show hire- and vacancy-level specifications as they are subject to selection bias and add little additional information beyond the firm-level specification.
- We drop outcomes for which the data quality is insufficient. This mostly affects hire-level outcomes, where we struggled to get adequate data on variables such as the ethnicity and religion of new hires (as well as variables derived from these). In addition, this list includes respondents' knowledge about prices of formal employee search methods.

- We include some outcomes that we did not pre-specify in the online appendix (e.g., number of employees at endline and share of white-collar employees at endline).
- We do not display all pre-specified heterogeneity analyses to simplify the presentation of results. We show a subset of this heterogeneity in Table [A14](#).

Figure E2: Vacancy search and type decision if white-collar positions are optimal for informal search



Notes: **Figure E2 displays the choice of search channel and vacancy type for firms in the $\pi(wc, formal)$ – $\pi(nwc, formal)$ space.** It assumes that informal search for non-white-collar positions has lower returns than informal search for white-collar positions ($\pi(nwc, informal) < \pi(wc, informal)$). Red lines indicate what happens when formal search costs are reduced. Firms that fall in the area above the dashed lines post white-collar vacancies (areas A, B, D, and E). Firms that fall below the dashed line post non-white-collar vacancies (areas C and F).